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“Berechnen Sie den Grenzwert” (0/0)	§5.3 Taylor-plug, beats l’Hospital	“konvergiert (f_n) gleichm.?”	§7.2 sup-norm recipe	“Beweisen Sie ... (Vorlesung)”	§11, ranked by likelihood
“Grenzwert einer Folge”	§3.2, or §3.3 if recursive	“punktweiser Grenzwert”	§7.1 freeze x , $n \rightarrow \infty$	“Welche Aussagen sind korrekt?”	§12.1 true/false scan index
“Konvergiert die Reihe?”	§4.2 decision tree	“gleichmäßig stetig?”	§6.5 recipe + case split	“Untersuchen Sie ...” (9 P)	§12.3 model write-ups
“Konvergenzradius / welche x ”	§5.1 radius recipe	“lim und $\int$ vertauschen?”	§7.3 interchange table	“Häufungspunkt / limsup”	§3.4 subsequence split
“Bestimmen Sie $\int f dx$ ”	§9.1 which method	“Riemann-integrierbar?”	§9.7 criteria list	“differenzierbar in x_0 ?”	§8.1 + Landau o/O
“allgemeine Lösung der DGL”	§10.1 which method	“uneigentliches Integral?”	§9.8 p -test, both ends	“Fixpunkt / eind. Nullstelle”	§6.4 IVT + monotonicity
“Kurvendiskussion / Extrema”	§8.4 full recipe	“Fehler abschätzen (Taylor)”	§5.2 remainder bound	“Ableitungstabelle benutzen”	§13.1 ! read this first

1 Foundations: sup/inf, induction, estimates, vectors

1.1 Quantifiers: read them, negate them

▶ RECIPE: negate any definition mechanically

Swap $\forall \leftrightarrow \exists$ left to right, keep the order, negate the final inequality ($<$ becomes $\geq$).

- $a_n \not\rightarrow a$: $\exists \varepsilon > 0 \forall N \exists n \geq N : |a_n - a| \geq \varepsilon$.
- not Cauchy**: $\exists \varepsilon > 0 \forall N \exists m, n \geq N : |a_n - a_m| \geq \varepsilon$.
- f **not continuous** at x_0 : $\exists \varepsilon > 0 \forall \delta > 0 \exists x : |x - x_0| < \delta$ and $|f(x) - f(x_0)| \geq \varepsilon$.
- f **not uniformly continuous**: $\exists \varepsilon > 0 \forall \delta > 0 \exists x, y : |x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon$.
- $f_n \not\rightarrow f$: $\exists \varepsilon > 0 \forall N \exists n \geq N \exists x : |f_n(x) - f(x)| \geq \varepsilon$.

In practice the last two become: build x_n, y_n resp. x_n and show the gap does not vanish.

TRAP: order of quantifiers is the whole content

$\forall x \exists \delta$ (continuity: δ may depend on x) versus $\exists \delta \forall x$ (uniform: one δ for all x). Same words, different theorem. Whenever a statement feels obvious, check whether you read the quantifiers in the right order.

1.2 sup / inf / max / min (Supremum, Infimum)

DEF: the four words

$M \subseteq \mathbb{R}$ nonempty. **upper bound** s : $x \leq s \forall x \in M$. $\sup M$ = least upper bound. $\max M = \sup M$ *only if* $\sup M \in M$. Mirrored for inf / min. **Completeness axiom**: every nonempty M bounded above has $\sup M \in \mathbb{R}$. This is the one property $\mathbb{Q}$ lacks and the engine behind IVT, Bolzano–Weierstrass and monotone convergence.

FACT: ε -characterisation (use it in every sup proof)

$s = \sup M \iff$ (i) $x \leq s \forall x \in M$, and (ii) $\forall \varepsilon > 0 \exists x \in M : x > s - \varepsilon$. For a function: $\sup_{x \in D} f(x)$ is the sup of the image set $f(D)$; it is a max iff attained.

▶ RECIPE: find sup / inf and prove it

- Guess s (the largest/smallest value the formula can approach).
- Bound**: show $x \leq s$ for all $x \in M$ by algebra.
- Least**: given $\varepsilon > 0$ produce $x \in M$ with $x > s - \varepsilon$ (usually choose $n > 1/\varepsilon$).
- Attained? $s \in M \Rightarrow \max = s$; otherwise max does not exist.

FACT: Archimedes and density

Archimedean: $\forall x \in \mathbb{R} \exists n \in \mathbb{N} : n > x$; equivalently $\frac{1}{n} \rightarrow 0$. $\mathbb{N}$ is unbounded in $\mathbb{R}$.
 $\mathbb{Q}$ **dense**: between any $a < b$ there is a rational *and* an irrational. Used to build the Dirichlet counterexamples.

TRAP: sup/inf in multiple choice

- sup exists as a *real number* only for a bounded-above set. Unbounded $\Rightarrow$ write $\sup = +\infty$, i.e. no real sup.
- min M need **not** be an accumulation point of the sequence: $x_0 = 0$, $x_n = 1 + \frac{1}{n}$ attains its min once.
- $\inf M = \sup M \Rightarrow M$ is a single point $\Rightarrow$ the sequence is constant $\Rightarrow$ converges. **T**
- monotone *and bounded* $\Rightarrow$ converges to sup (incr.) / inf (decr.). Monotone alone may run to $\pm\infty$.

1.3 Absolute value and inequalities

▶ RECIPE: solve $|\cdot|$ inequalities

$|x - a| < r \iff x \in (a - r, a + r)$; $|x - a| > r \iff x < a - r$ or $x > a + r$.
 $|f(x)| < g(x) \iff -g(x) < f(x) < g(x)$ (and $g > 0$).
General: split at every zero of the inside, solve on each piece, intersect with that piece.

FACT: the inequalities you actually use

- Triangle**: $|a + b| \leq |a| + |b|$; **reverse**: $||a| - |b|| \leq |a - b|$.
 - Bernoulli**: $(1 + x)^n \geq 1 + nx$ for $x \geq -1$.
 - AM–GM**: $\sqrt{ab} \leq \frac{a+b}{2}$ ($a, b \geq 0$).
 - Young**: $ab \leq \frac{a^2}{2} + \frac{b^2}{2}$, sharper $ab \leq \frac{\varepsilon a^2}{2} + \frac{b^2}{2\varepsilon}$.
 - Cauchy–Schwarz**: $|\langle x, y \rangle| \leq \|x\| \|y\|$.
- Always available**: $|\sin t| \leq 1$, $|\sin t| \leq |t|$ (strict for $t \neq 0$), $|\cos t| \leq 1$, $e^t \geq 1 + t$, $\ln t \leq t - 1$.

▶ RECIPE: estimate without panic

Target is always $|\text{expr}| \leq$ (something $\rightarrow 0$).

- Numerator: make it *bigger* (drop negative terms, bound $\sin / \cos$ by 1).
- Denominator: make it *smaller* (drop positive terms). Never the reverse.
- Roots: conjugate, $\sqrt{A} - \sqrt{B} = \frac{A-B}{\sqrt{A}+\sqrt{B}}$.
- Sums: bound every term by the largest, then multiply by the count.

1.4 Induction (vollständige Induktion)

▶ RECIPE: proof by induction

- Base**: verify $n = n_0$ explicitly.
 - Hypothesis**: assume $A(n)$ for one fixed $n \geq n_0$.
 - Step**: write the $n+1$ statement, isolate the n part, substitute the hypothesis, then estimate to the target. Say where you used it.
- Helpers**: $\sum_{k=1}^n k = \frac{n(n+1)}{2}$, $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum_{k=0}^n q^k = \frac{1-q^{n+1}}{1-q}$, $(1+x)^n \geq 1+nx$.

1.5 Vectors, inner product, norms in $\mathbb{R}^n$

DEF: inner product, norm, distance

$\langle x, y \rangle = x \cdot y = \sum_{i=1}^n x_i y_i$; $\|x\| = \sqrt{\langle x, x \rangle} = \sqrt{\sum x_i^2}$; $d(x, y) = \|x - y\| = \sqrt{\sum (x_i - y_i)^2}$.

FACT: Cauchy–Schwarz and the norm axioms

$|\langle x, y \rangle| \leq \|x\| \|y\|$, with **equality iff** x, y are linearly dependent. This is exactly what makes the triangle inequality true.
A **norm** satisfies: $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$; $\|cx\| = |c| \|x\|$; $\|x + y\| \leq \|x\| + \|y\|$.
Hence $\|x - z\| \leq \|x - y\| + \|y - z\|$ for all x, y, z .

2 Complex numbers and polynomials

2.1 Forms, conversion, arithmetic

DEF: the two forms

$z = a + ib = re^{i\varphi} = r(\cos \varphi + i \sin \varphi)$; $\bar{z} = a - ib$, $|z| = r = \sqrt{a^2 + b^2}$, $z\bar{z} = |z|^2$.
Re $z = \frac{z+\bar{z}}{2}$, Im $z = \frac{z-\bar{z}}{2i}$. Euler: $e^{i\varphi} = \cos \varphi + i \sin \varphi$, $e^{i\pi} = -1$, $e^{2\pi i} = 1$.

▶ RECIPE: convert both ways

- Cartesian $\rightarrow$ polar**: $r = \sqrt{a^2 + b^2}$; $\varphi = \arctan \frac{b}{a}$ and then add π if $a < 0$ (subtract 2π to land in $(-\pi, \pi]$). ! always check the quadrant; arctan alone loses half the plane.
- $a = 0$: $\varphi = \frac{\pi}{2}$ if $b > 0$, $-\frac{\pi}{2}$ if $b < 0$.
- Polar $\rightarrow$ Cartesian**: $a = r \cos \varphi$, $b = r \sin \varphi$.

FACT: arithmetic rules

Multiply/divide in polar: $r_1 e^{i\varphi_1} \cdot r_2 e^{i\varphi_2} = r_1 r_2 e^{i(\varphi_1 + \varphi_2)}$, $\frac{z}{w} = \frac{r_z}{r_w} e^{i(\varphi_z - \varphi_w)}$.
Divide in Cartesian: multiply by $\frac{\bar{w}}{w}$: $\frac{z}{w} = \frac{z\bar{w}}{|w|^2}$.
Conjugate: $\overline{z + w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, $\overline{z/w} = \bar{z}/\bar{w}$, $\bar{\bar{z}} = z$, $z \in \mathbb{R} \iff z = \bar{z}$, $|z| = |\bar{z}|$.
de Moivre: $(\cos \varphi + i \sin \varphi)^n = \cos(n\varphi) + i \sin(n\varphi)$.

▶ RECIPE: use Euler to derive trig identities on the spot

$\cos x = \frac{e^{ix} + e^{-ix}}{2}$, $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$. Multiply the exponentials, collect.
Ex. $\cos^2 x = \left(\frac{e^{ix} + e^{-ix}}{2} \right)^2 = \frac{e^{2ix} + 2 + e^{-2ix}}{4} = \frac{1 + \cos 2x}{2}$. Same trick gives every double/half-angle formula, so you never have to memorise them.

2.2 Roots and equations

▶ RECIPE: solve $z^n = w$ (n -th roots)

- Write $w = \rho e^{i\psi}$.
 - $z_k = \rho^{1/n} e^{i(\psi + 2\pi k)/n}$ for $k = 0, 1, \dots, n-1$.
 - Exactly n roots, all on the circle of radius $\rho^{1/n}$, spaced by $\frac{2\pi}{n}$.
- Roots of unity**: $z^n = 1 \Rightarrow z_k = e^{2\pi i k/n}$; their sum is 0 for $n \geq 2$ and their product is $(-1)^{n+1}$.

▶ RECIPE: quadratic with complex data

$az^2 + bz + c = 0 \Rightarrow z = \frac{-b \pm \sqrt{D}}{2a}$, $D = b^2 - 4ac$.

- Real coefficients, $D < 0$: $z = -\frac{b}{2a} \pm i \frac{\sqrt{|D|}}{2a}$ (a conjugate pair).
- Complex D : get $\sqrt{D}$ with the n -th-root recipe at $n = 2$ (two roots, take either).

FACT: complex log and general powers (if asked)

$\log z = \ln |z| + i(\arg z + 2\pi k)$, $k \in \mathbb{Z}$ — multivalued; the *principal* branch takes $\arg z \in (-\pi, \pi]$.
 $z^w := e^{w \log z}$. ! $\sqrt{zw} = \sqrt{z}\sqrt{w}$ and $\log(zw) = \log z + \log w$ can fail by $2\pi i$ off the principal branch.

2.3 Polynomials: what is guaranteed

FACT: Fundamental theorem of algebra

p of degree $n \geq 1$ with *any* complex coefficients has **exactly n roots in $\mathbb{C}$ counted with multiplicity**, and factors as $p(z) = a_n \prod_j (z - z_j)^{m_j}$ with $\sum m_j = n$.

TRAP: the three classic multiple-choice errors

- “ n roots, all *distinct*” — **F**, only with multiplicity (z^5 has one root of multiplicity 5).
- “complex roots come in conjugate pairs” — **F** in general, **T only if all coefficients are real**. Degree 5 with complex coefficients need not have any real root.
- “odd degree $\Rightarrow$ a real root” — **T only for real coefficients** (IVT on p , which changes sign at $\pm\infty$).

▶ RECIPE: factor a polynomial over $\mathbb{R}$ / over $\mathbb{C}$

1. Guess a root among the divisors of the constant term (over $\mathbb{Z}$), or spot $z = \pm 1$.
2. Polynomial division by $(z - z_0)$, repeat on the quotient.
3. Over $\mathbb{C}$: keep going until all factors are linear.
4. Over $\mathbb{R}$: pair each non-real root with its conjugate to get an irreducible quadratic $z^2 - 2\operatorname{Re}(z_0)z + |z_0|^2$. Real factorisation = linear factors $\times$ irreducible quadratics only.

3 Sequences (a_n)

3.1 Definitions you must be able to write

DEF: convergence, divergence, boundedness

$a_n \rightarrow a : \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |a_n - a| < \varepsilon$.
 $a_n \rightarrow \infty : \iff \forall K > 0 \exists N \forall n \geq N : a_n > K$ (divergence to ∞ , not convergence).
bounded: $\exists C : |a_n| \leq C \forall n$. **monotone incr.**: $a_{n+1} \geq a_n \forall n$.
explicit $a_n = f(n)$ **vs recursive** $a_{n+1} = f(a_n)$ — the treatment is completely different, see §3.2 vs §3.3.

FACT: the implication chain (know which arrows reverse)

convergent $\Rightarrow$ Cauchy $\Rightarrow$ bounded $\Rightarrow$ has a convergent subsequence (Bolzano–Weierstrass).
 Only reversal that holds: **Cauchy** $\Leftrightarrow$ **convergent** in $\mathbb{R}$ (completeness). Also monotone + bounded $\Leftrightarrow$ convergent for *monotone sequences*.

3.2 Limit of an explicitly given a_n

▶ RECIPE: compute $\lim a_n$; match the shape

1. **Rational in n** : divide top and bottom by the highest power in the denominator. deg top < bottom $\Rightarrow 0$; $= \Rightarrow$ ratio of leading coefficients; $> \Rightarrow \pm\infty$.
2. **Roots**, $\infty - \infty$: multiply by the conjugate.
3. **n -th roots, n in the exponent**: write $a_n = e^{\ln a_n}$ and take the limit in the exponent.
4. $(1 + \frac{a}{n})^{dn}$: $\rightarrow e^{cd}$.
5. **Factorials**, n^n : use the growth order $\ln n \ll n^\alpha \ll b^n \ll n! \ll n^n$ ($\alpha > 0, b > 1$).
6. $(-1)^n$ or $\sin / \cos$ **times a null sequence**: squeeze $-|b_n| \leq a_n \leq |b_n|$.
7. $\frac{1}{n} \sum_{k \leq n} f(k/n)$: that is a Riemann sum, see §9.6.
8. **Nothing fits**: treat n as a real x and use Taylor / l'Hospital (§5.3, §8.5).

FACT: tools with names — cite them, it earns marks

Limit laws: sums, products, quotients (denominator $\neq 0$) — only if *both* limits exist finitely.
Squeeze (Sandwich): $b_n \leq a_n \leq c_n$ and $b_n, c_n \rightarrow L \Rightarrow a_n \rightarrow L$.
Monotone convergence (Monotoniekriterium): monotone + bounded $\Rightarrow$ convergent, limit = sup resp. inf of the term set.
Continuity transfer: $a_n \rightarrow a$ and f continuous at $a \Rightarrow f(a_n) \rightarrow f(a)$.
Ratio criterion for sequences: $\left| \frac{a_{n+1}}{a_n} \right| \rightarrow q < 1 \Rightarrow a_n \rightarrow 0$ geometrically.

EXAMPLE: the limits that keep coming back

- $\frac{3n^2 - n}{2n^2 + 5} = \frac{3 - \frac{1}{n}}{2 + \frac{5}{n^2}} \rightarrow \frac{3}{2}$.
- $\sqrt{n^2 + n} - n = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \rightarrow \frac{1}{2}$.
- $\sqrt[n]{n} = e^{\frac{\ln n}{n}} \rightarrow 1$; $\sqrt[n]{a} \rightarrow 1$ ($a > 0$); $\sqrt[n]{2^n + 3^n} \rightarrow 3$ (pull out the biggest).
- $\left(1 + \frac{3}{n}\right)^{2n} = \left[\left(1 + \frac{3}{n}\right)^n\right]^2 \rightarrow e^6$.
- $\frac{n!}{n^n} \leq \frac{1}{n} \rightarrow 0$; $\frac{2^n}{n!} \rightarrow 0$ (ratio $\frac{2}{n+1} \leq \frac{1}{2}$ for $n \geq 3$).
- $\frac{(-1)^n}{n}$: $|a_n| \leq \frac{1}{n} \rightarrow 0 \Rightarrow \rightarrow 0$. Alternating by itself means nothing.

▶ RECIPE: prove $a_n \rightarrow a$ by ε - N

1. Compute $|a_n - a|$, simplify to one fraction.
2. Over-estimate to a clean $\frac{C}{n^k}$ (numerator bigger, denominator smaller).
3. Solve $\frac{C}{n^k} < \varepsilon \Rightarrow$ take $N := \lceil (C/\varepsilon)^{1/k} \rceil + 1$.
4. Write it out: “Let $\varepsilon > 0$, choose N as above; then $\forall n \geq N$: $|a_n - a| \leq \frac{C}{n^k} < \varepsilon$.” $\square$

FACT: estimates worth remembering

$n! \geq 2^{n-1}$; $\left(\frac{n}{3}\right)^n \leq n! \leq n^n$; **Stirling** $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$, so $\sqrt[n]{n!} \sim \frac{n}{e} \rightarrow \infty$.
 $\sum_{k=1}^n \frac{1}{k} \approx \ln n + \gamma$ (grows, unbounded); $\left(1 + \frac{1}{n}\right)^n \uparrow e$, $\left(1 + \frac{1}{n}\right)^{n+1} \downarrow e$.

3.3 Recursive sequences $x_{n+1} = f(x_n)$

▶ RECIPE: full treatment (a standard 4–9 point task)

1. **Candidate limit**: solve the fixed-point equation $L = f(L)$. Keep every solution.
2. **Bounded**: induction — find an interval I with $x_0 \in I$ and $f(I) \subseteq I$, conclude $x_n \in I$ for all n .
3. **Monotone**: the sign of $f(x) - x$ on I gives the direction; if f is increasing, that sign never flips, so (x_n) is monotone.
4. **Conclude**: monotone + bounded $\Rightarrow$ converges. Then $L = \lim x_{n+1} = \lim f(x_n) = f(L)$ by continuity of f ; pick the root of $L = f(L)$ lying in I .

FACT: contraction shortcut (often much faster)

If $|f'(x)| \leq q < 1$ on I then by the MVT $|x_{n+1} - L| = |f(x_n) - f(L)| \leq q|x_n - L|$, hence $|x_n - L| \leq q^n |x_0 - L| \rightarrow 0$.
 $\Rightarrow$ converges to the **unique** fixed point, for *every* starting point $x_0 \in I$. No monotonicity needed.

EXAMPLE: $x_{n+1} = \frac{\sin x_n}{1 + x_n^2}$, $x_0 \in (0, 1)$

$f(x) = \frac{\sin x}{1 + x^2}$. **1)** $f(0) = 0$; for $x \neq 0$, $|\sin x| < |x|$ and $1 + x^2 \geq 1$ give $|f(x)| < |x| \Rightarrow$ no other fixed point. **2)** $x_0 \in (0, 1) \Rightarrow \sin x_0 > 0 \Rightarrow$ all $x_n > 0$, and $x_{n+1} < x_n \Rightarrow$ strictly decreasing, bounded below by 0. **3)** Monotone + bounded $\Rightarrow$ converges to $L \geq 0$ with $L = f(L) \Rightarrow L = 0$.

3.4 Accumulation points, limsup, liminf

DEF: accumulation point (Häufungspunkt)

h is an accumulation point of $(a_n) : \iff$ every ε -ball around h contains a_n for **infinitely many** $n \iff$ some subsequence converges to h .
 $\limsup a_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} a_k$ = largest accumulation point; $\liminf$ = smallest.

▶ RECIPE: find all accumulation points

1. Split by the periodic part: $(-1)^n \Rightarrow n = 2k$ and $n = 2k + 1$; $\cos \frac{n\pi}{3} \Rightarrow$ period 6; $\sin n \Rightarrow$ dense, every point of $[-1, 1]$.
2. Take the limit of each subsequence. That set is exactly the set of accumulation points.
3. $\limsup = \max$, $\liminf = \min$ of that set.

Ex. $a_n = (-1)^n \frac{n}{n+1} \Rightarrow \{-1, +1\}$, $\limsup = 1$, $\liminf = -1$, diverges.

FACT: limsup / liminf algebra

Always defined in $[-\infty, \infty]$, no hypotheses needed. $\liminf a_n \leq \limsup a_n$.
 • a_n converges $\iff$ bounded and $\liminf = \limsup$ (then all three coincide).
 • $\limsup(a_n + b_n) \leq \limsup a_n + \limsup b_n$ — **!** only $\leq$, equality can fail ($a_n = (-1)^n$, $b_n = -(-1)^n$).
 • $\liminf(a_n + b_n) \geq \liminf a_n + \liminf b_n$; $\liminf(-a_n) = -\limsup a_n$.
 • If $b_n \rightarrow b$ exists then $\limsup(a_n + b_n) = \limsup a_n + b$ (a convergent summand splits off exactly).

TRAP: monotone sequences and accumulation points

A monotone sequence is either bounded ($\Rightarrow$ converges, **exactly** one accumulation point) or unbounded ($\Rightarrow \rightarrow \pm\infty$, **no** accumulation point).
 So “a monotone sequence may have no accumulation point” is **T**, and “at least one” / “at least two” are both **F**. Bolzano–Weierstrass only promises one for *bounded* sequences.

FACT: nested intervals (Intervallschachtelung)

$I_1 \supseteq I_2 \supseteq \dots$ closed, bounded, $|I_n| \rightarrow 0 \Rightarrow \bigcap I_n$ is exactly one point. Equivalent to completeness; it is the engine of the bisection proof of Bolzano–Weierstrass.

3.5 Cauchy sequences

DEF: Cauchy

$\forall \varepsilon > 0 \exists N \forall m, n \geq N : |a_n - a_m| < \varepsilon$. In $\mathbb{R}$: **Cauchy** $\iff$ **convergent**, so this proves convergence *without knowing the limit*.

TRAP: the near-miss

“ $|a_{n+1} - a_n| \rightarrow 0$ ” is **not** enough: $a_n = \sqrt{n}$ and $a_n = \sum_{k \leq n} \frac{1}{k}$ both satisfy it and diverge. You need control of $|a_n - a_m|$ for *all* $m, n \geq N$, not just neighbours.

▶ RECIPE: show (a_n) is Cauchy

Standard route: show $|a_{n+1} - a_n| \leq Cq^n$ with $q < 1$. Then for $m > n$ $|a_m - a_n| \leq \sum_{k=n}^{m-1} Cq^k \leq \frac{Cq^n}{1-q} \rightarrow 0$, so Cauchy, so convergent.

4 Series $\sum a_n$

4.1 Setup

DEF: series, absolute vs conditional

$\sum_{n=1}^\infty a_n := \lim_{N \rightarrow \infty} s_N$ with $s_N = \sum_{n=1}^N a_n$ — a series is the sequence of partial sums.
absolutely convergent: $\sum |a_n| < \infty$. **abs. conv.** $\Rightarrow$ **conv.**, not conversely ($\sum \frac{(-1)^n}{n}$).
conditionally convergent: converges but not absolutely.
 Convergence is unaffected by finitely many terms or by the starting index.

4.2 Decision tree: does $\sum a_n$ converge?

▶ RECIPE: run these in order, stop at the first hit

1. $a_n \not\rightarrow 0 ? \Rightarrow$ **diverges**, done (*Trivialkriterium*). Free, always check first.
2. **Known shape?** geometric, p -series, log- p — read the answer off the box below.
3. **Factorials** or $c^n ? \Rightarrow$ **ratio test**.
4. **n -th powers, n in the exponent?** $\Rightarrow$ **root test** (stronger than ratio).
5. **Resembles a known series?** $\Rightarrow$ **comparison** on leading powers.
6. **Alternating?** $\Rightarrow$ **Leibniz**.
7. $a_n = f(n)$ with $f \geq 0$ decreasing? $\Rightarrow$ **integral test**.
8. **Telescoping?** $a_n = b_n - b_{n+1} \Rightarrow s_N = b_1 - b_{N+1}$.

FACT: the benchmark series (memorise these four lines)

- $\sum_{n \geq 0} q^n$ conv. $\iff |q| < 1$, value $\frac{1}{1-q}$; $\sum_{n \geq 1} nq^{n-1} = \frac{1}{(1-q)^2}$.
- $\sum \frac{1}{n^p}$ conv. $\iff p > 1$. ($p = 1$ harmonic, diverges.)
- $\sum_{n \geq 2} \frac{1}{n \ln^p n}$ conv. $\iff p > 1$.
- $\sum \frac{1}{n!} = e$; $\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} = \ln 2$; $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$; $\sum \frac{1}{n(n+1)} = 1$.

FACT: ratio and root test, stated exactly

Ratio (Quotientenkriterium): $q = \lim \left| \frac{a_{n+1}}{a_n} \right|$. $q < 1$ abs. conv.; $q > 1$ div.; $q = 1$ **no information**.
Root (Wurzelkriterium): $q = \limsup \sqrt[n]{|a_n|}$, same verdicts.
! Both give $q = 1$ for $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$, which behave differently — at $q = 1$ you must switch method.

FACT: comparison, Leibniz, integral test

Comparison (Majorante/Minorante): $0 \leq a_n \leq b_n$, $\sum b_n < \infty \Rightarrow \sum a_n < \infty$; $a_n \geq b_n \geq 0$, $\sum b_n = \infty \Rightarrow \sum a_n = \infty$. Compare leading powers only: $\frac{3n^2 + n}{n^4 - 1} \sim \frac{3}{n^2} \Rightarrow$ conv.
Limit comparison: $\frac{a_n}{b_n} \rightarrow c \in (0, \infty) \Rightarrow$ same behaviour.
Leibniz: $a_n = (-1)^n b_n$ with $b_n \geq 0$ **monotonically** decreasing to 0 $\Rightarrow$ converges, and $|s - s_N| \leq b_{N+1}$.
Integral test: $f \geq 0$ decreasing $\Rightarrow \int_1^\infty f dx \leq \sum_{n \geq 1} f(n) \leq f(1) + \int_1^\infty f dx$; sum and integral converge together.

TRAP: test traps

- Leibniz needs *monotone* b_n : $b_n = \frac{2+(-1)^n}{n} \rightarrow 0$ but is not monotone.
- Comparison needs $a_n \geq 0$; with signs, apply it to $|a_n|$ (that proves *absolute* convergence only).
- $\sum a_n$ conv. and $\sum b_n$ conv. $\Rightarrow \sum (a_n + b_n)$ conv. **T**, but $\sum a_n b_n$ conv. **F** ($a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$).
- “ s_N bounded $\Rightarrow$ converges” **F** unless $a_n \geq 0$ (then s_N is monotone).

EXAMPLE: run the tree on five series

- $\sum \frac{n^2}{2^n}$: $c^n \Rightarrow$ ratio $\frac{(n+1)^2}{2^{n+1}} \rightarrow \frac{1}{2} < 1 \Rightarrow$ **conv.**
- $\sum \frac{n!}{n^n}$: ratio $= \left(\frac{n}{n+1}\right)^n \rightarrow \frac{1}{e} < 1 \Rightarrow$ **conv.**
- $\sum \left(\frac{n}{2n+1}\right)^n$: root $= \frac{n}{2n+1} \rightarrow \frac{1}{2} < 1 \Rightarrow$ **conv.**
- $\sum \frac{2n+3}{n^2+7}$: behaves like $\frac{2}{n} \Rightarrow$ **div.**
- $\sum \frac{(-1)^n}{\sqrt{n}}$: Leibniz $\Rightarrow$ conv.; $\sum \frac{1}{\sqrt{n}}$ div. $\Rightarrow$ **conditionally conv.**

EXAMPLE: telescoping via partial fractions

$$\sum_{n \geq 1} \frac{1}{n(n+2)}: \quad \text{split} \quad \frac{1}{n(n+2)} = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+2} \right). \quad s_N = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{N+1} - \frac{1}{N+2} \right) \rightarrow \frac{3}{4}.$$

Recipe: partial-fraction it, write out the first three and last two terms, see what survives.

4.3 Rearrangement, products, mixed-sign tests

FACT: rearrangement and Cauchy product

- Riemann rearrangement:** conditionally convergent $\Rightarrow$ a reordering can be made to converge to *any* prescribed value, or to $\pm\infty$. Order matters.
- Dirichlet:** absolutely convergent $\Rightarrow$ every reordering gives the same sum.
- Cauchy product** $c_n = \sum_{k=0}^n a_k b_{n-k}$: if at least one factor converges absolutely then $\sum c_n = (\sum a_n)(\sum b_n)$ (Mertens).

FACT: Abel / Dirichlet test (for $\sum a_n b_n$ with signs)

If $b_n \downarrow 0$ monotonically and the partial sums of $\sum a_n$ are *bounded*, then $\sum a_n b_n$ converges.

This is what makes $\sum \frac{\sin n}{n}$ and $\sum \frac{\cos n}{n}$ converge although neither is absolutely convergent, and it is the discrete twin of the integration-by-parts argument for $\int_1^\infty \frac{\cos t}{t} dt$.

5 Power series and Taylor series

5.1 Radius of convergence

► RECIPE: “for which x does $\sum a_n(x-x_0)^n$ converge?”

- Read off the **centre** x_0 (in $(x-2)^n$: $x_0 = 2$) and the **coefficient** a_n = everything without $(x-x_0)^n$.
- Radius** (Cauchy–Hadamard): $\frac{1}{R} = \lim \left| \frac{a_{n+1}}{a_n} \right|$ for factorials, or $\frac{1}{R} = \limsup \sqrt[n]{|a_n|}$ for powers. Limit $0 \Rightarrow R = \infty$; limit $\infty \Rightarrow R = 0$.
- Absolute convergence on $|x-x_0| < R$, divergence on $|x-x_0| > R$.
- Endpoints** $x = x_0 \pm R$ **one at a time** — plug in and use §4.2 (usually Leibniz or p -series). The two ends can behave differently.
- Answer as an interval, e.g. $[x_0 - R, x_0 + R)$.

EXAMPLE: $\sum_{n \geq 1} \frac{n^n (x-2)^n}{(2n)!}$

$a_n = \frac{n^n}{(2n)!}$; factorial $\Rightarrow$ ratio test.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{n^n} = \frac{(n+1) \left(1 + \frac{1}{n}\right)^n}{(2n+1)(2n+2)} \rightarrow 0,$$

since the numerator grows like $e n$ and the denominator like $4n^2$. So $\frac{1}{R} = 0$, $R = \infty$: **converges for every** $x \in \mathbb{R}$.

FACT: what a power series gives you for free

On $|x-x_0| < R$ the sum is C^∞ and you may **differentiate and integrate termwise**; R is unchanged. Coefficients are unique: $a_n = \frac{f^{(n)}(x_0)}{n!}$.

Convergence is *uniform* on every closed $[x_0 - r, x_0 + r]$ with $r < R$ (Weierstrass M -test), **!** not necessarily on the whole open disc.

5.2 Taylor formula and error bounds

FACT: Taylor with Lagrange remainder

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k}_{T_n(x)} + R_n(x), \quad R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-x_0)^{n+1} \text{ for}$$

some ξ between x_0 and x ; also $R_n = o(|x-x_0|^n)$.

Error bound: $|R_n(x)| \leq \frac{M}{(n+1)!} |x-x_0|^{n+1}$ with $M = \max |f^{(n+1)}|$ on the interval between x_0 and x .

► RECIPE: approximate a value and bound the error

- Pick x_0 near x where f and its derivatives are known exactly (0, 1, $\frac{\pi}{6}$, ...).
- Write T_n and evaluate.
- Bound $M = \max |f^{(n+1)}|$ on the interval (for sin, cos: $M = 1$ always; for e^x : the right endpoint).
- If an accuracy ε is prescribed, solve $\frac{M}{(n+1)!} |x-x_0|^{n+1} < \varepsilon$ for n .

Ex. $e^{0.1}$, $n = 3$: $T_3 = 1 + 0.1 + 0.005 + \frac{0.001}{6} \approx 1.10517$; $M < 1.2 \Rightarrow |R_3| \leq \frac{1.2}{24} 10^{-4} = 5 \cdot 10^{-6}$.

DEF: linear / tangent approximation

$f(x) \approx f(x_0) + f'(x_0)(x-x_0)$ — the $n = 1$ Taylor polynomial, i.e. the tangent line. Error $\leq \frac{\max |f''|}{2} (x-x_0)^2$.

This is also the *definition* of differentiability: the tangent is the unique linear function whose error is $o(|x-x_0|)$.

5.3 Limits by Taylor: the fastest tool you have

► RECIPE: limits $x \rightarrow 0$ by plugging in series

- Look at the denominator's power x^m . You need every series to order m — one order further if the leading terms cancel.
- Insert the standard series (§13.3), expand, cancel.
- The surviving lowest-order term is the answer; everything else is $o(x^m)$.

Order bookkeeping: $\sin x = x + O(x^3)$, so $\frac{\sin x}{x} = 1 + O(x^2)$. Track $O(x^k)$ explicitly and you cannot lose a term.

EXAMPLE: $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$

$$\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots$$

Numerator $= \frac{x^5}{120} + O(x^7) \Rightarrow$ limit $= \frac{1}{120}$.

l'Hospital would need **five** differentiations. Don't.

EXAMPLE: more Taylor limits, one line each

- $\frac{1 - \cos x}{x^2} = \frac{x^2/2 + O(x^4)}{x^2} \rightarrow \frac{1}{2}$.
- $\frac{e^x - 1 - x}{x^2} = \frac{x^2/2 + O(x^3)}{x^2} \rightarrow \frac{1}{2}$.
- $\frac{\tan x - x}{x^3} \rightarrow \frac{1}{3}$ (since $\tan x = x + \frac{x^3}{3} + O(x^5)$).
- $\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x} = \frac{x^3/6}{x^2} \rightarrow 0$.
- $\frac{\ln(1+x) - x}{x^2} \rightarrow -\frac{1}{2}$.

5.4 Building new series from old ones

► RECIPE: never recompute derivatives — transform instead

- Substitute:** $e^{-x^2} = \sum \frac{(-1)^n x^{2n}}{n!}$; $\frac{1}{1+x^2} = \sum (-1)^n x^{2n}$.
- Differentiate termwise:** $\frac{1}{(1-x)^2} = \sum_{n \geq 1} n x^{n-1}$.
- Integrate termwise:** $\arctan x = \int \frac{dx}{1+x^2} = \sum \frac{(-1)^n x^{2n+1}}{2n+1}$; $\ln(1+x) = \int \frac{dx}{1+x}$.
- Multiply:** collect by total degree (Cauchy product). $e^x \sin x = x + x^2 + \frac{x^3}{3} + O(x^4)$.
- Divide:** long division of the series, or multiply by the geometric expansion of the denominator.
- Shift centre:** to expand at $x_0 \neq 0$, substitute $t = x - x_0$ and expand in t .

FACT: ready-made compositions

$$\ln(1+x^2) = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \dots; \quad \sin(x^2) = x^2 - \frac{x^6}{6} + \dots; \quad e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \dots;$$
$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - \dots; \quad \sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \dots; \quad \frac{1}{\sqrt{1-x}} = 1 + \frac{x}{2} + \frac{3x^2}{8} + \dots$$

6 Functions: limits, continuity, IVT/EVT

6.1 Basic properties, inverse

► RECIPE: injective / surjective / bijective / inverse

- Injective:** $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. Shortcut: *strictly* monotone $\Rightarrow$ injective (show $f' > 0$ or $f' < 0$ throughout).
- Surjective onto Y :** solve $f(x) = y$ for arbitrary $y \in Y$, or get $\text{im } f = Y$ from IVT plus the limits at the ends of the domain.
- Bijective** = both $\Rightarrow f^{-1}$ exists; compute it by solving $y = f(x)$ for x , then swap names.
- Regularity of f^{-1} :** f continuous and strictly monotone on an interval $\Rightarrow f^{-1}$ continuous and strictly monotone; and differentiable with $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$ whenever $f'(x_0) \neq 0$.

Also: even $f(-x) = f(x)$, odd $f(-x) = -f(x)$, periodic $f(x+T) = f(x)$, bounded $|f| \leq C$, restriction $f|_A$.

FACT: monotone functions are well behaved

Monotone on an interval $\Rightarrow$ at most *countably many* discontinuities, all of them **jumps**; both one-sided limits exist everywhere; Riemann integrable on every $[a, b]$. It can still fail to be continuous, and it is injective only if *strictly* monotone.

6.2 Limits of functions

DEF: ε - δ , one-sided, at infinity

$\lim_{x \rightarrow x_0} f(x) = L : \iff \forall \varepsilon > 0 \exists \delta > 0 : 0 < |x-x_0| < \delta \Rightarrow |f(x)-L| < \varepsilon$.

Limit exists $\iff \lim_{x \rightarrow x_0^-} f = \lim_{x \rightarrow x_0^+} f$, both finite.

$\lim_{x \rightarrow \infty} f = L : \iff \forall \varepsilon \exists K : x > K \Rightarrow |f(x)-L| < \varepsilon$.

► RECIPE: compute $\lim_{x \rightarrow x_0} f(x)$

- Plug in x_0 . Nothing breaks $\Rightarrow$ done, by continuity.
- $\frac{0}{0} \Rightarrow$ factor and cancel, expand roots by the conjugate, else Taylor (§5.3) or l'Hospital (§8.5).
- $\frac{c \neq 0}{0} \Rightarrow \pm\infty$; get the sign from each side separately, so the answer may exist only one-sided.
- $x \rightarrow \pm\infty$ rational $\Rightarrow$ divide by the highest power in the denominator. **!** for $x \rightarrow -\infty$, $\sqrt{x^2} = |x| = -x$.
- $|x|$, sgn, floor, piecewise $\Rightarrow$ **always** both one-sided limits.
- Bounded $\times$ null $\Rightarrow$ squeeze: $x \sin \frac{1}{x} \rightarrow 0$, while $\sin \frac{1}{x}$ has no limit at all.

6.3 Continuity

DEF: continuity and the sequential criterion

f continuous at $x_0 : \iff \forall \varepsilon > 0 \exists \delta > 0 : |x-x_0| < \delta \Rightarrow |f(x)-f(x_0)| < \varepsilon$

$\iff \lim_{x \rightarrow x_0} f(x) = f(x_0) \iff$ (Heine) $x_n \rightarrow x_0$ implies $f(x_n) \rightarrow f(x_0)$ for every such sequence.

► RECIPE: prove or disprove continuity at x_0

Prove: if f is built from $+$, $-$, $\times$, $\div$ (denominator $\neq 0$) and o of continuous standard functions, say exactly that — one line. Otherwise ε - δ : bound $|f(x) - f(x_0)| \leq C|x-x_0|$ and take $\delta = \varepsilon/C$.

Disprove: give *two* sequences $x_n \rightarrow x_0$, $y_n \rightarrow x_0$ with $\lim f(x_n) \neq \lim f(y_n)$ (Heine). Classics: $x_n = \frac{1}{2\pi n}$ vs $y_n = \frac{1}{\frac{\pi}{2} + 2\pi n}$ for $\sin \frac{1}{x}$; rationals vs irrationals for Dirichlet.

FACT: types of discontinuity (Unstetigkeitsstellen)

removable (*hebbbar*): both one-sided limits exist and agree, but $\neq f(x_0)$ — redefine and it is continuous.

jump (*Sprung*): both exist and differ. This is the monotone-function case.

essential / oscillating: at least one one-sided limit fails to exist ($\sin \frac{1}{x}$, $\frac{1}{x}$).

► **RECIPE:** fit a parameter: “determine a, b so that f is continuous / differentiable”

1. Continuity at the seam x_0 : $\lim_{x \rightarrow x_0^-} f = \lim_{x \rightarrow x_0^+} f = f(x_0) \Rightarrow$ one equation.
2. Differentiability: additionally $\lim_{x \rightarrow x_0^-} f' = \lim_{x \rightarrow x_0^+} f' \Rightarrow$ second equation.
3. Solve the (usually linear) system. Two unknowns need exactly these two conditions.

6.4 The three theorems on $[a, b]$ — and when to reach for which

FACT: cite these verbatim, hypotheses included

$f : [a, b] \rightarrow \mathbb{R}$ **continuous**, $[a, b]$ **compact**:

- **IVT** (*Zwischenwertsatz*): c between $f(a)$ and $f(b) \Rightarrow \exists x \in [a, b] : f(x) = c$. For: a zero, a solution, a fixed point exists.
- **Extreme value theorem** (*Satz vom Max/Min, Weierstrass*): f is bounded and *attains* max and min. For: “the maximum is attained”, bounding.
- **Heine**: continuous on a compact interval $\Rightarrow$ *uniformly* continuous. For: *uniform continuity* for free.

! All three fail on open or unbounded intervals: $\frac{1}{x}$ on $(0, 1]$ is continuous, unbounded, attains no max, is not uniformly continuous.

► **RECIPE:** show “ $f(x) = g(x)$ has a solution” / “ f has a zero”

1. Set $h := f - g$ (fixed point $f(x) = x$: take $h(x) := f(x) - x$). State: h continuous as a difference of continuous functions.
2. Find two points of opposite sign — evaluate at convenient points, or use the signs of $\lim_{x \rightarrow \pm\infty} h$ and pick large $|x|$.
3. IVT $\Rightarrow \exists x^*$ with $h(x^*) = 0$. **Existence done.**
4. **Uniqueness:** h strictly monotone ($h' > 0$ or $h' < 0$ everywhere) $\Rightarrow$ at most one zero. Or Rolle: two zeros would force $h'(\xi) = 0$, contradiction.

EXAMPLE: $e^x = p(x)$ — the standard multiple choice

- “IVT always gives a solution” **F**: $p \equiv -1$ has none, since $e^x > 0$.
- p of *odd* degree $\Rightarrow$ always at least one solution ($e^x - p(x)$ changes sign at $\pm\infty$).
- “at most finitely many solutions for every p ” **T**: $e^x - p(x)$ is analytic and not identically 0, so its zeros are isolated and finite in number on $\mathbb{R}$ here.
- “for every N there is a p_N with $\geq N$ solutions” **T**: take p_N close to e^x on a long interval and let it wobble across.

6.5 Uniform continuity (gleichmässige Stetigkeit)

DEF: the difference that matters

continuous: $\forall x_0 \forall \varepsilon \exists \delta(\varepsilon, x_0)$. **uniformly continuous:** $\forall \varepsilon \exists \delta(\varepsilon) \forall x, y : |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$.

One δ for the whole domain. unif. cont. $\Rightarrow$ cont., never the converse.

Chain: Lipschitz $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous.

► **RECIPE:** “is f uniformly continuous?”

1. **Compact domain** $[a, b]$, f **continuous**? $\Rightarrow$ YES by Heine. One line, done.
2. f' **bounded**, $|f'| \leq L?$ $\Rightarrow$ MVT gives $|f(x) - f(y)| \leq L|x - y|$, Lipschitz $\Rightarrow$ YES with $\delta = \varepsilon/L$. *The workhorse on unbounded domains.*
3. **Continuous on $\mathbb{R}$ with finite limits at $\pm\infty$?** $\Rightarrow$ YES (compact in the middle, nearly constant outside).
4. **Suspect NO?** Produce x_n, y_n with $|x_n - y_n| \rightarrow 0$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0 > 0$.
5. **Parameter that can degenerate** (e.g. $n = 0$ vs $n \geq 1$) $\Rightarrow$ case distinction, treat the degenerate case on its own.

! Unbounded f' does **not** prove failure: $\sqrt{x}$ on $[0, 1]$ has $f' \rightarrow \infty$ and is uniformly continuous (compact).

EXAMPLE: uniformly continuous: $f_n(x) = \frac{x}{1 + nx^2}$ on $\mathbb{R}$

$n = 0$: $f_0(x) = x$ is Lipschitz with $L = 1$.

$n \geq 1$: $f'_n(x) = \frac{1 - nx^2}{(1 + nx^2)^2}$, so $|f'_n(x)| \leq \frac{1 + nx^2}{(1 + nx^2)^2} = \frac{1}{1 + nx^2} \leq 1$.

Bounded derivative $\Rightarrow$ Lipschitz with $L = 1 \Rightarrow$ **uniformly continuous for every n** , with $\delta = \varepsilon$.

EXAMPLE: not uniformly continuous: the two patterns

$f(x) = x^2$ **on $\mathbb{R}$** : $x_n = n, y_n = n + \frac{1}{n} \Rightarrow |x_n - y_n| = \frac{1}{n} \rightarrow 0$ but $|f(y_n) - f(x_n)| = 2 + \frac{1}{n^2} \geq 2$. $\Rightarrow$ fails for $\varepsilon = 2$.

$f(x) = \frac{1}{x}$ **on $(0, 1)$** : $x_n = \frac{1}{n}, y_n = \frac{1}{2n} \Rightarrow$ difference $\rightarrow 0$ but $|f(y_n) - f(x_n)| = n \rightarrow \infty$.

$\sin \frac{1}{x}$ **on $(0, 1)$** : pick x_n, y_n at consecutive max/min of the oscillation $\Rightarrow$ gap 2.

7 Sequences of functions (f_n)

7.1 Pointwise limit: always compute this first

DEF: pointwise vs uniform, side by side

pointwise: $\forall x \forall \varepsilon \exists N(\varepsilon, x) \forall n \geq N : |f_n(x) - f(x)| < \varepsilon$ — N may depend on x .

uniform: $\forall \varepsilon \exists N(\varepsilon) \forall n \geq N \forall x : |f_n(x) - f(x)| < \varepsilon$, i.e.

$$\|f_n - f\|_\infty := \sup_{x \in D} |f_n(x) - f(x)| \xrightarrow{n \rightarrow \infty} 0.$$

One N for all x at once. Uniform $\Rightarrow$ pointwise, never the converse.

► **RECIPE:** find the pointwise limit $f(x) = \lim_n f_n(x)$

1. **Freeze x** , treat it as a constant, let $n \rightarrow \infty$ with §3.2.
2. **Case distinction in x** wherever the answer changes: usually $x = 0$ vs $x \neq 0$, or $|x| < 1$ vs $|x| = 1$ vs $|x| > 1$.
3. Write f piecewise. If f is discontinuous while all f_n are continuous, convergence cannot be uniform — that alone answers part (c).

7.2 Uniform convergence $f_n \rightrightarrows f$

► **RECIPE:** decide uniform convergence (the standard 3-point task)

1. Get the pointwise limit f first — uniform convergence can only be towards it.
2. Set $g_n(x) := |f_n(x) - f(x)|$ and compute $M_n := \sup_x g_n(x)$: differentiate in x with n fixed, solve $g'_n(x) = 0$, substitute the critical point back. Check the boundary and $|x| \rightarrow \infty$ too.
3. $M_n \rightarrow 0 \Rightarrow$ **uniform**. $M_n \not\rightarrow 0 \Rightarrow$ **pointwise only**.

► **RECIPE:** two shortcuts that give a complete disproof

- **Exhibit one sequence x_n** with $|f_n(x_n) - f(x_n)| \geq \varepsilon_0 > 0$ (take $x_n =$ the maximiser). One such sequence is a full proof of failure — you do not need the exact sup.
- **All f_n continuous but f is not** $\Rightarrow$ not uniform, by the contrapositive of the uniform limit theorem.

EXAMPLE: $f_n(x) = \frac{x}{1 + nx^2}$ on $\mathbb{R}$: **uniform**

Pointwise $f \equiv 0$, so $M_n = \sup_x \left| \frac{x}{1 + nx^2} \right|$.

$f'_n(x) = \frac{1 - nx^2}{(1 + nx^2)^2} = 0 \Rightarrow x = \pm \frac{1}{\sqrt{n}}$, and $f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1/\sqrt{n}}{2} = \frac{1}{2\sqrt{n}}$; $f_n \rightarrow 0$ as $|x| \rightarrow \infty$.

So $M_n = \frac{1}{2\sqrt{n}} \rightarrow 0 \Rightarrow$ **uniformly to 0 on all of $\mathbb{R}$** .

TRAP: sup bounds that do and do not block uniformity

- $\sup_x |f_n(x)| \geq \frac{1}{2n}$ does **not** prevent $f_n \rightrightarrows 0 - \frac{1}{2n} \rightarrow 0$. Only a lower bound that *does not vanish* blocks it.
- The max being attained at a *different* point x_n for each n is irrelevant. Only the *value* M_n counts.
- $f_n \rightarrow f$ pointwise with all f_n and f continuous still need not be uniform ($nx e^{-nx^2}$ on $[0, 1]$).

TABLE: gallery: memorise this table, it answers most (f_n) questions

f_n (domain)	ptwise f	$\ f_n - f\ _\infty$	unif.? f swaps?
x^n on $[0, 1]$	0 on $[0, 1]$, 1 at 1	1	F T
x^n on $[0, r]$, $r < 1$	0	r^n	T T
$\frac{x}{1 + nx^2}$ on $\mathbb{R}$	0	$\frac{1}{2\sqrt{n}}$	T T
$\frac{\sin(nx)}{n}$ on $\mathbb{R}$	0	$\frac{1}{n}$	T T
$nx e^{-nx^2}$ on $[0, 1]$	0	$\sqrt{\frac{n}{2}} e^{-1/2}$	F F
$n^2 x e^{-nx}$ on $[0, 1]$	0	$\frac{n}{2}$	F F
$\arctan(nx)$ on $\mathbb{R}$	$\frac{\pi}{2} \operatorname{sgn} x$	$\frac{\pi}{2}$	F n/a
$\frac{x^n}{1 + x^n}$ on $[0, 1]$	0, $\frac{1}{2}$ at 1	$\frac{1}{2}$	F T

Read it as: a discontinuous limit is *never* uniform. Uniformity is sufficient but not necessary for swapping f : row 1 swaps without it ($\int_0^1 x^n = \frac{1}{n+1} \rightarrow 0$), rows 5–6 do not ($\int_0^1 nx e^{-nx^2} \rightarrow \frac{1}{2}$, $\int_0^1 n^2 x e^{-nx} \rightarrow 1$, both $\neq 0$).

FACT: uniform Cauchy criterion and the M -test

Uniform Cauchy: $f_n \rightrightarrows$ some limit $\iff \forall \varepsilon \exists N \forall m, n \geq N : \|f_n - f_m\|_\infty < \varepsilon$. Use it when you do not know f .

Weierstrass M -test (for $\sum f_n$): $|f_n(x)| \leq M_n \forall x \in D$ and $\sum M_n < \infty \Rightarrow \sum f_n$ converges **absolutely and uniformly** on D . This is how power series get uniform convergence on $[-r, r]$, $r < R$.

Dini: D compact, f_n continuous, f continuous, and $f_n \rightarrow f$ pointwise *monotonically* in $n \Rightarrow$ convergence is automatically uniform. The only cheap converse there is.

7.3 Interchanging limits with $\int$ and $\frac{d}{dx}$

TABLE: what you may swap, and the exact price

Swap	Hypothesis that makes it legal	Counterexample if missing
$\lim_n \int_a^b f_n = \int_a^b \lim_n f_n$	$f_n \rightrightarrows f$ on the <i>compact</i> $[a, b]$	$n^2 x e^{-nx}$ on $[0, 1]$: $\rightarrow 1 \neq 0$
$\lim_n f'_n = (\lim_n f_n)'$	$f'_n \rightrightarrows g$ uniformly <i>and</i> $f_n(x_0)$ converges for one x_0	$\frac{\sin(nx)}{n} \rightrightarrows 0$ but $f'_n = \cos(nx)$ diverges
$\sum_n \int f_n = \int \sum_n f_n$	$\sum_n f_n$ uniformly conv. on $[a, b]$ (M -test)	same shape as row 1
$\frac{d}{dx} \sum_n f_n = \sum_n f'_n$	$\sum f'_n$ uniformly convergent	holds free inside a power series' radius
$\lim_{x \rightarrow x_0} \lim_n f_n = \lim_n \lim_{x \rightarrow x_0} f_n$	$f_n \rightrightarrows f$ (then f is continuous)	x^n on $[0, 1]$ at $x_0 = 1$

FACT: what uniform convergence buys you

f_n continuous on D and $f_n \rightrightarrows f$:

- f is **continuous** — the *Uniform Limit Theorem*. Pointwise alone gives nothing.
- On $[a, b]$: f is Riemann integrable and $\int_a^b f_n \rightarrow \int_a^b f$.
- f_n bounded uniformly $\Rightarrow f$ bounded.
- **! Derivatives do not follow.** Uniform convergence of f_n says nothing about f'_n ; you must assume f'_n converges uniformly.

8 Derivatives and their applications

8.1 Differentiability

DEF: three equivalent formulations

$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ (**two-sided**, finite)

$\iff \exists K \in \mathbb{R} : f(x) = f(x_0) + K(x - x_0) + o(|x - x_0|)$ as $x \rightarrow x_0$, and then $K = f'(x_0)$.

diff. $\Rightarrow$ **continuous**; continuous $\not\Rightarrow$ diff. ($|x|$ at 0).

TRAP: Landau o / O — the classic multiple-choice trap

$f = O(g)$: $|f| \leq C|g|$ near x_0 (*same order or smaller*). $f = o(g)$: $f/g \rightarrow 0$ (*strictly smaller*).

- $f(x) - f(x_0) - K(x - x_0) = o(|x - x_0|) \Rightarrow$ **differentiable T**. Note $(x - x_0)$, *not* Kx : written with Kx the statement is meaningless.
- $f(x) - f(x_0) = O(|x - x_0|) \Rightarrow$ only Lipschitz near $x_0 \Rightarrow$ **not** differentiable **F** ($|x|$ satisfies it!).
- only $h \rightarrow 0^+$ exists $\Rightarrow$ only *right*-differentiable $\Rightarrow$ not differentiable **F**.
- “for an arbitrary polynomial p ” $\Rightarrow$ useless **F**; p must be exactly the linear Taylor part.

Algebra: $o(x^n) \pm o(x^n) = o(x^n)$, $x^m \cdot o(x^n) = o(x^{m+n})$, $o(x^n) = O(x^n)$ but not conversely.

► RECIPE: piecewise function: differentiable at the seam x_0 ?

- Continuity first: $\lim_{x \rightarrow x_0^-} f = \lim_{x \rightarrow x_0^+} f = f(x_0)$. Fails $\Rightarrow$ not differentiable, stop.
- Compare the one-sided *difference quotients* (not just $\lim f'$). Equal $\Rightarrow$ differentiable, with that value.
- For C^1 additionally $\lim_{x \rightarrow x_0^-} f' = \lim_{x \rightarrow x_0^+} f'$.

Ex. $f(x) = x^2 \sin \frac{1}{x}$ ($x \neq 0$), $f(0) = 0$: $f'(0) = \lim h \sin \frac{1}{h} = 0$ exists, but f' is not continuous at 0 $\Rightarrow$ differentiable, *not* C^1 .

8.2 Rules — and how not to misuse them

FACT: the rules

$(f \pm g)' = f' \pm g'$; $(fg)' = f'g + fg'$; $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$;
chain $(g \circ f)'(x) = g'(f(x))f'(x)$; **inverse** $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$, $y_0 = f(x_0)$;
Leibniz $(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$;
logarithmic for $u(x)^{v(x)}$: write $u^v = e^{v \ln u} \Rightarrow (u^v)' = u^v (v' \ln u + \frac{vu'}{u})$.

► RECIPE: implicit differentiation

Given $F(x, y) = 0$ with $y = y(x)$: differentiate both sides in x treating y as a function, so every y contributes a factor y' by the chain rule, then solve for y' .

Ex. $x^2 + y^2 = 1 \Rightarrow 2x + 2yy' = 0 \Rightarrow y' = -\frac{x}{y}$.

TRAP: using the derivative/antiderivative table correctly

The table (§13.1) lists $F(x)$ for the **bare** argument x . If the argument is $ax + b$:

$$\int f(ax + b) dx = \frac{1}{a} F(ax + b) + C \quad (\text{divide by the inner derivative})$$

! Valid only for linear inner functions. Anything else: substitute properly.

Worked trap: $\int \frac{1}{2 \cos^2(\frac{1}{2}x)} dx$. Table: $\int \frac{du}{\cos^2 u} = \tan u$. With $u = \frac{x}{2}$, $dx = 2 du$: $\int \frac{2 du}{2 \cos^2 u} = \tan u = \tan \frac{x}{2} + C$ — **not** $2 \tan \frac{x}{2}$. Always differentiate your answer. Five seconds, catches every chain-rule slip.

8.3 Mean value theorems (Mittelwertsätze)

FACT: Fermat / Rolle / Lagrange / Cauchy

- Fermat:** x_0 an *interior* local extremum, f differentiable there $\Rightarrow f'(x_0) = 0$. Converse false (x^3).
- Rolle:** $f \in C[a, b]$, diff. on (a, b) , $f(a) = f(b) \Rightarrow \exists \xi \in (a, b)$: $f'(\xi) = 0$.
- MVT (Lagrange):** same hypotheses $\Rightarrow \exists \xi \in (a, b)$: $f'(\xi) = \frac{f(b) - f(a)}{b - a}$.
- Cauchy MVT:** $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$ — this is what proves l'Hospital.

► RECIPE: the four things MVT is for

- Prove** $|f(x) - f(y)| \leq L|x - y|$: MVT plus a bound $|f'| \leq L$. (Route to Lipschitz $\Rightarrow$ uniform continuity.)
- Prove f constant:** $f' \equiv 0$ on an interval $\Rightarrow$ constant.
- Count zeros:** n zeros of $f \Rightarrow$ at least $n - 1$ zeros of f' (Rolle between consecutive ones). Contrapositive: f' has $\leq k$ zeros $\Rightarrow f$ has $\leq k + 1$.
- Compare two functions:** set $g = f - h$, show $g(a) = 0$ and $g' > 0 \Rightarrow f > h$ on (a, ∞) .

8.4 Curve discussion (Kurvendiskussion) and extrema

► RECIPE: find and classify all extrema of f on I

- Domain**, plus every point where f is not differentiable (kinks, poles) and the endpoints of I — all are candidates.
- Critical points:** solve $f'(x) = 0$.
- Classify:** $f''(x_0) > 0 \Rightarrow$ local **min**; $f''(x_0) < 0 \Rightarrow$ local **max**; $f''(x_0) = 0 \Rightarrow$ inconclusive, use the *sign change* of f' ($- \rightarrow +$ min, $+ \rightarrow -$ max, no change $\Rightarrow$ saddle).
- Global on $[a, b]$:** evaluate f at all critical points *and* at a, b ; largest wins (it exists, by Weierstrass).
- Global on open/unbounded I :** also take $\lim_{x \rightarrow \partial I} f$; if a limit beats every candidate, the extremum is *not attained*.
- Monotonicity:** sign table of f' . **Convexity:** sign table of f'' ; a sign change of f'' is an **inflection point** (*Wendepunkt*).
- Asymptotes:** $m = \lim \frac{f(x)}{x}$, $c = \lim(f(x) - mx)$; then $y = mx + c$.

EXAMPLE: full discussion of $f(x) = \frac{x}{1 + x^2}$

Domain $\mathbb{R}$, odd. $f' = \frac{1 - x^2}{(1 + x^2)^2} \Rightarrow$ critical $x = \pm 1$. $f'' = \frac{2x(x^2 - 3)}{(1 + x^2)^3} \Rightarrow f''(1) < 0$: **max** $f(1) = \frac{1}{2}$; $f''(-1) > 0$: **min** $f(-1) = -\frac{1}{2}$. Inflection at $x = 0, \pm\sqrt{3}$. Increasing on $[-1, 1]$, decreasing outside. $f \rightarrow 0$ at $\pm\infty \Rightarrow$ asymptote $y = 0$. Global max/min $= \pm\frac{1}{2}$, both attained.

FACT: convexity (konvex) — exam favourite

f convex on $I \iff f'' \geq 0 \iff f'$ increasing $\iff f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \iff$ the graph lies *above* every tangent.

- Convex and $f'(x_0) = 0$ at an interior point $\Rightarrow x_0$ is a **global minimum T**.
- A convex differentiable f cannot have several *isolated* critical points — the critical set is an interval **F**.
- Its global *maximum* on $[a, b]$ is always attained at a **boundary point T**.
- “the global minimum can only be on the boundary” **F** — it may well be interior.

► RECIPE: prove an inequality with convexity

- Tangent-line trick:** f convex $\Rightarrow f(x) \geq f(a) + f'(a)(x - a)$ for all x . Choosing a good a proves most one-variable inequalities in one line, e.g. $e^x \geq 1 + x$ ($a = 0$), $\ln x \leq x - 1$ ($f = -\ln$, $a = 1$).
- Jensen:** f convex $\Rightarrow f(\frac{1}{n} \sum x_i) \leq \frac{1}{n} \sum f(x_i)$. With $f = -\ln$ this is AM–GM.

8.5 l'Hospital (Bernoulli–de l'Hospital)

► RECIPE: use it safely

- Check the form first:** only $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$. Applying it unchecked loses marks.
- Differentiate *numerator and denominator separately* — **!** not the quotient rule.
- Repeat while the form persists.
- Verify the new limit exists. If it does not, l'Hospital states **nothing** about the original.

Reduce other forms first: $0 \cdot \infty \Rightarrow \frac{f}{1/g}$; $\infty - \infty \Rightarrow$ common denominator or conjugate; $1^\infty, 0^0, \infty^0 \Rightarrow u^v = e^{v \ln u}$, work in the exponent, exponentiate at the end.

TRAP: when l'Hospital fails

- $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x} = 1$, but l'Hospital gives $1 + \cos x$, which has no limit $\Rightarrow$ inapplicable. Divide by x instead.
- $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{1 + x^2}}$: l'Hospital cycles forever. Divide by x .
- Repeated use that never resolves $(\frac{e^x}{e^x})$ patterns $\Rightarrow$ switch to Taylor or algebra.

Rule of thumb: at $x \rightarrow 0$ prefer Taylor (§5.3); it is faster and cannot mislead you.

EXAMPLE: indeterminate forms, one line each

- $\lim_{x \rightarrow 0^+} x \ln x = \lim \frac{\ln x}{1/x} \stackrel{\infty/\infty}{=} \lim \frac{1/x}{-1/x^2} = \lim(-x) = 0$.
- $\lim_{x \rightarrow 0^+} x^x = e^{\lim x \ln x} = e^0 = 1$.
- $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^{\lim x \ln(1 + a/x)} = e^a$.
- $\lim_{x \rightarrow 0^+} \frac{\ln x}{\cot x} = \frac{-\infty}{\infty} \Rightarrow$ l'Hospital $\Rightarrow \lim \frac{1/x}{-1/\sin^2 x} = -\lim \frac{\sin^2 x}{x} = 0$.

9 Integration

9.1 Which method? Match the integrand

► RECIPE: take the first match, then go to that subsection

- Already in the table (§13.1)?** $\Rightarrow$ read it off. Inner function $ax + b \Rightarrow$ multiply by $\frac{1}{a}$, nothing else may be inner.
- The inner derivative appears as a factor**, $\int g(f(x))f'(x) dx \Rightarrow$ **substitution** $u = f(x)$, §9.2.
- Product of two unrelated types** (poly $\times e^x/\sin/\ln/\arctan$) $\Rightarrow$ **by parts**, §9.3.
- Rational function** $\frac{p}{q} \Rightarrow$ division if $\deg p \geq \deg q$, then **partial fractions**, §9.4.
- Rational in sin, cos** $\Rightarrow$ half-angle identities first, else $t = \tan \frac{x}{2}$, §9.5.
- $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2} \Rightarrow$ **trig substitution** $x = a \sin \theta / a \tan \theta / a \sec \theta$.
- $\sqrt[n]{ax + b} \Rightarrow$ substitute $u = \sqrt[n]{ax + b}$; the root disappears completely.
- $\sin^m x \cos^n x \Rightarrow$ one odd power: peel a factor off and substitute the other. Both even: half-angle.
- $\sin^n x$, $\cos^n x$, $x^n e^x$ with large $n \Rightarrow$ **reduction formula**, §9.3.
- Nothing works** $\Rightarrow$ symmetry (§9.6), or the answer legitimately contains $\int \frac{\sin x}{x} dx$, which is non-elementary and allowed in a multiple-choice option.

Always differentiate your result at the end. Under time pressure it is the only check that matters.

9.2 Substitution

► RECIPE: step by step

- Choose $u = g(x)$, usually the inner function of a composition.
- $du = g'(x) dx \Rightarrow$ solve for dx and make *every* x disappear.
- Definite:** convert the bounds $a \mapsto g(a)$, $b \mapsto g(b)$ — then do *not* substitute back.
- Indefinite:** integrate in u , then **substitute** $u = g(x)$ **back**.
- ! Differentiate the result:** the chain-rule factor $g'(x)$ must cancel. This is where most marks are lost.

Reverse direction ($x = \varphi(t)$, φ bijective and C^1): $\int f(x) dx = \int f(\varphi(t))\varphi'(t) dt$, back via $t = \varphi^{-1}(x)$.

FACT: the three substitution patterns worth memorising

$$\int \frac{f'}{f} dx = \ln |f| + C; \quad \int f^n f' dx = \frac{f^{n+1}}{n+1} + C; \quad \int f' e^f dx = e^f + C.$$

Ex. $\int \frac{x}{1+x^2} dx$: $u = 1 + x^2$, $du = 2x dx \Rightarrow \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln(1 + x^2) + C$.

EXAMPLE: trig substitution done properly

$\int \sqrt{1 - x^2} dx$: put $x = \sin \theta$, $dx = \cos \theta d\theta \Rightarrow \int \cos^2 \theta d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4}$.
Back-substitute with $\theta = \arcsin x$ and $\sin 2\theta = 2 \sin \theta \cos \theta = 2x \sqrt{1 - x^2}$:

$$\int \sqrt{1 - x^2} dx = \frac{1}{2} \arcsin x + \frac{1}{2} x \sqrt{1 - x^2} + C.$$

! Back-substitution needs the identity, not just $\theta = \arcsin x$.

9.3 Integration by parts (partielle Integration)

► RECIPE: $\int uv' dx = uv - \int u'v dx$

- Choose $u =$ the factor that gets *simpler* when differentiated. Priority **LIATE**: Log, Inverse trig, Algebraic, Trig, Exponential — earlier in the list becomes u .
- $v' =$ the rest; integrate it to get v (no $+C$ here).
- Apply, repeat if the new integral is still a product ($x^2 e^x$: twice).
- $\int \ln x dx$, $\int \arctan x dx$: take $v' = 1 \Rightarrow x \ln x - x$, $x \arctan x - \frac{1}{2} \ln(1 + x^2)$.

► **RECIPE: cycle trick (when the integral comes back)**

$I = \int e^x \sin x \, dx$: two rounds of parts return $I = -e^x \cos x + e^x \sin x - I$. Treat I as an unknown and **solve algebraically**: $2I = e^x(\sin x - \cos x) \Rightarrow I = \frac{e^x(\sin x - \cos x)}{2} + C$.

Same for $\int e^{ax} \cos(bx) \, dx$ and $\int \sec^3 x \, dx$.

FACT: reduction formulas

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

$$\int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

$$\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx; \quad \int (\ln x)^n \, dx = x(\ln x)^n - n \int (\ln x)^{n-1} \, dx$$

Iterate down to $n = 0$ or 1 . Definite version on $[0, \frac{\pi}{2}]$: $\int_0^{\pi/2} \sin^n = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2}$.

EXAMPLE: multiple choice: “ $\int x \ln x \sin x \, dx = \dots$ ”

Do **not** integrate. **Differentiate the four options** and compare with the integrand: 20 seconds instead of 5 minutes.

Check of $-x \ln x \cos x + \ln x \sin x + \sin x - \int \frac{\sin x}{x} \, dx$:
 $-(\ln x + 1) \cos x + x \ln x \sin x + \frac{\sin x}{x} + \ln x \cos x + \cos x - \frac{\sin x}{x} = x \ln x \sin x$ ✓

9.4 Partial fractions (Partialbruchzerlegung)

► **RECIPE: $\int \frac{p(x)}{q(x)} \, dx$**

- deg $p \geq \deg q \Rightarrow$ polynomial division first.
- Factor q completely over $\mathbb{R}$: linear factors and irreducible quadratics only (§2.3).
- Ansatz: $(x - a) \rightarrow \frac{A}{x-a}$; $(x - a)^k \rightarrow \frac{A_1}{x-a} + \dots + \frac{A_k}{(x-a)^k}$; irreducible $x^2 + bx + c \rightarrow \frac{Ax+B}{x^2+bx+c}$.
- Multiply out and compare coefficients, or plug in the roots (**cover-up** is fastest for simple linear factors).
- Integrate: $\int \frac{A}{x-a} = A \ln|x - a|$; $\int \frac{A}{(x-a)^k} = \frac{A}{(1-k)(x-a)^{k-1}}$; for $\frac{Ax+B}{x^2+bx+c}$ split off $\frac{A}{2} \cdot \frac{2x+b}{x^2+bx+c}$ (gives $\ln$) and complete the square on the rest (gives arctan).

Ex. $\int \frac{dx}{x^2-1} = \int \left(\frac{1/2}{x-1} - \frac{1/2}{x+1} \right) dx = \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$.

9.5 Rational in sin and cos

► **RECIPE: half-angle first, Weierstrass only if needed**

Try first: $1 + \cos x = 2 \cos^2 \frac{x}{2}$, $1 - \cos x = 2 \sin^2 \frac{x}{2}$, $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$.

Weierstrass $t = \tan \frac{x}{2}$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2}{1+t^2} dt \Rightarrow$ rational in $t \Rightarrow$ §9.4.

EXAMPLE: $\int \frac{dx}{1+\cos x}$ (and its twin)

$$1 + \cos x = 2 \cos^2 \frac{x}{2} \Rightarrow \int \frac{dx}{2 \cos^2 \frac{x}{2}} \stackrel{u=x/2}{=} \int \frac{2 \, du}{2 \cos^2 u} = \tan u = \boxed{\tan \frac{x}{2} + C}$$

Check: $\frac{d}{dx} \tan \frac{x}{2} = \frac{1}{2 \cos^2 \frac{x}{2}} = \frac{1}{1+\cos x}$ ✓

Twin: $\int \frac{dx}{1-\cos x} = -\cot \frac{x}{2} + C$.

9.6 Definite integrals, FTC, symmetry

FACT: Fundamental theorem (Hauptsatz)

f continuous on $[a, b]$, $F(x) := \int_a^x f(t) \, dt \Rightarrow F$ is differentiable and $F' = f$.

Newton–Leibniz: $\int_a^b f = F(b) - F(a)$ for any antiderivative F .

Variable bounds (chain rule!): $\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) \, dt = f(v(x))v'(x) - f(u(x))u'(x)$.

MVT for integrals: f continuous $\Rightarrow \exists \xi \in [a, b]$ with $\int_a^b f = f(\xi)(b - a)$.

FACT: free simplifications

- f **odd** on a symmetric interval $\Rightarrow \int_{-a}^a f = 0$; f **even** $\Rightarrow = 2 \int_0^a f$.
- $\int_a^b = -\int_b^a$; $\int_a^b = \int_a^c + \int_c^b$; $| \int f | \leq \int |f|$; $f \leq g \Rightarrow \int f \leq \int g$.
- $|\int_a^b f| \leq (b - a) \max |f|$ (the standard estimate).
- Period T : $\int_a^{a+T} f$ does not depend on a .
- $\int_0^{2\pi} \sin^2 = \int_0^{2\pi} \cos^2 = \pi$; $\int_0^\pi \sin x \, dx = 2$; $\int_0^{\pi/2} \sin^2 = \frac{\pi}{4}$.

► **RECIPE: limit of a sum = integral (Riemann sum trick)**

A limit of the shape $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right)$ equals $\int_0^1 f(x) \, dx$.

- Factor out $\frac{1}{n}$ (the width). If it is not there, the trick does not apply.
- Write the rest as a function of $\frac{k}{n}$ alone $\Rightarrow$ that is f .
- General interval: $\frac{b-a}{n} \sum f\left(a + k \frac{b-a}{n}\right) \rightarrow \int_a^b f$.

Ex. $\lim \sum_{k=1}^n \frac{n}{n^2+k^2} = \lim \frac{1}{n} \sum \frac{1}{1+(k/n)^2} = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$.

Ex. $\lim \frac{1}{n} \sum_{k=1}^n \ln \frac{k}{n} = \int_0^1 \ln x \, dx = -1$.

► **RECIPE: integrals with $|\cdot|$, max, or pieces**

- Find every point where the sign or the active branch changes (zeros of the inside).
- Split there, drop the $|\cdot|$ with the correct sign on each piece.
- Add up. !Never pull $|\cdot|$ outside the integral.

Ex. $\int_0^2 |x - 1| \, dx = \int_0^1 (1 - x) \, dx + \int_1^2 (x - 1) \, dx = \frac{1}{2} + \frac{1}{2} = 1$.

9.7 Riemann integrability (Riemann-integrierbar)

DEF: Darboux definition

Partition $P : a = x_0 < \dots < x_n = b$; $L(f, P) = \sum_i (\inf_{[x_{i-1}, x_i]} f) \Delta x_i$, $U(f, P) = \sum_i (\sup_{[x_{i-1}, x_i]} f) \Delta x_i$.

f (bounded) is integrable $\iff \sup_P L = \inf_P U \iff \forall \varepsilon > 0 \exists P : U(f, P) - L(f, P) < \varepsilon$.

To compute from the definition: take the equidistant P , evaluate the two sums, let $n \rightarrow \infty$.

FACT: criteria — and the false friends

- T** continuous on $[a, b] \Rightarrow$ integrable. Monotone on $[a, b] \Rightarrow$ integrable. Differentiable $\Rightarrow$ continuous $\Rightarrow$ integrable.
- T** bounded with *finitely* (even countably) many discontinuities $\Rightarrow$ integrable. Lebesgue criterion: bounded and the discontinuity set is a null set $\iff$ integrable.
- F** “must be continuous everywhere” — jumps are fine.
- F** “integrable $\Rightarrow$ differentiable” — a step function is integrable and not even continuous.
- F** “finitely many function *values* $\Rightarrow$ integrable” — Dirichlet $1_{\mathbb{Q}}$ takes only 0, 1 and is not integrable.
- ! Riemann integrability requires **bounded** f ; unbounded means improper.

9.8 Improper integrals (uneigentliche Integrale)

► **RECIPE: does it converge?**

- Locate every problem point**: $\pm\infty$ and every point where f is unbounded. !Split so each piece has *exactly one* problem; *all* pieces must converge.
- Write each as a limit: $\int_a^\infty f = \lim_{R \rightarrow \infty} \int_a^R f$, $\int_a^b f = \lim_{\varepsilon \rightarrow 0^+} \int_a^{b-\varepsilon} f$.
- Compare with the benchmark below at that specific point.
- Limit comparison**: $\frac{f(x)}{g(x)} \rightarrow c \in (0, \infty) \Rightarrow$ same behaviour; replace f by its leading behaviour there.
- Sign changes**: test absolute convergence first; if that fails, oscillating integrals may still converge conditionally (parts / Dirichlet).

TABLE: benchmarks: the exponent flips meaning at the two ends

Integral	Converges iff
$\int_1^\infty x^{-p} \, dx$	$p > 1$ (needs <i>fast</i> decay at ∞)
$\int_1^\infty x^{-p} \, dx$	$p < 1$ (only a <i>mild</i> singularity at 0)
$\int_0^\infty \frac{dx}{x}$	$p > 1$
$\int_0^\infty \frac{x \ln p \, x}{x}$	$p > 1$
$\int_0^\infty e^{-\lambda x} \, dx = \frac{1}{\lambda}$	$\lambda > 0$
$\int_1^\infty \frac{\sin x}{x} \, dx$	converges, <i>conditionally</i> (parts)
$\int_0^1 \ln x \, dx = -1$	converges

EXAMPLE: inspect both ends before answering

- $\int_0^\infty \frac{\sin t}{t^3} \, dt$: at $t \rightarrow 0$, $\frac{\sin t}{t^3} \sim \frac{1}{t^2} \Rightarrow$ **diverges**. The problem is at 0, not at ∞ .
- $\int_0^\infty \frac{dt}{t^2+t^4}$: at $t \rightarrow 0$, $\sim \frac{1}{t^2} \Rightarrow$ **diverges**.
- $\int_1^\infty \frac{\cos^2 t}{t} \, dt$: $\cos^2 t = \frac{1+\cos 2t}{2}$ contains $\int \frac{dt}{2t} \Rightarrow$ **diverges**.
- $\int_1^\infty \frac{\cos t}{t} \, dt$: parts with $u = \frac{1}{t} \Rightarrow$ **converges** conditionally.

10 Differential equations (ODE)

10.1 Which method? Classify first

► **RECIPE: read the shape, then jump**

- $y' = f(x)g(y)$ (factorises) $\Rightarrow$ **separation**, §10.2.
- $y' + p(x)y = q(x)$ (linear, 1st order) $\Rightarrow$ **integrating factor**, §10.3.
- $y' + p(x)y = q(x)y^\alpha \Rightarrow$ **Bernoulli**, $z = y^{1-\alpha}$, §10.4.
- $y' = f\left(\frac{y}{x}\right) \Rightarrow u = \frac{y}{x}$, i.e. $y = ux$, $y' = u'x + u \Rightarrow$ separable.
- $y' = f(ax + by + c) \Rightarrow u = ax + by + c \Rightarrow$ separable.
- $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_0y = g(x)$, **constant** $a_i \Rightarrow$ characteristic polynomial (§10.5) + particular solution (§10.6).
- $y' = f(y)$ (**autonomous**) $\Rightarrow$ separable; the constant solutions are the zeros of f , and they decide the qualitative picture (§10.7).

Order of work, always: general solution first $\Rightarrow$ then insert the initial/boundary conditions $\Rightarrow$ then solve for the constants.

10.2 Separation of variables (Trennung der Variablen)

► **RECIPE: $y' = f(x)g(y)$**

- ! **First** solve $g(y_0) = 0$: each root gives a **constant solution** $y \equiv y_0$. Forgetting these costs marks and they are often the ones the initial condition needs.
- For $g(y) \neq 0$: $\frac{dy}{g(y)} = f(x) \, dx$, integrate both sides, one constant C on the right.
- Solve for y ; handle $|\cdot|$ and $\pm$ by absorbing into C ($e^C =: \tilde{C} > 0$, the sign extends it to $\tilde{C} \in \mathbb{R}$).
- Insert the initial condition, then state the interval of existence.

Ex. $y' = xy$: $\frac{dy}{y} = x \, dx \Rightarrow \ln|y| = \frac{x^2}{2} + C \Rightarrow y = \tilde{C}e^{x^2/2}$ (which includes $y \equiv 0$ at $\tilde{C} = 0$).

EXAMPLE: separable with an initial condition

$y' = y^2 \cos x$, $y(0) = 1$. Constant solution $y \equiv 0$ (does not meet $y(0) = 1$). Otherwise $-\frac{1}{y} = \sin x + C \Rightarrow y = \frac{-1}{\sin x + C}$; $y(0) = 1 \Rightarrow C = -1 \Rightarrow$

$$y = \frac{1}{1 - \sin x}, \text{ valid near } x = 0 \text{ until } \sin x = 1.$$

10.3 First order linear $y' + p(x)y = q(x)$

► **RECIPE: integrating factor (fastest, always works)**

- Normal form $y' + p(x)y = q(x)$ — the coefficient of y' must be 1.
- $\mu(x) := e^{\int p(x) \, dx}$ (no $+C$ needed).
- $(\mu y)' = \mu q \Rightarrow y = \frac{1}{\mu(x)} \left(\int \mu(x)q(x) \, dx + C \right)$.
- Insert the initial condition last.

Same thing as “variation of constants”: $y_h = Ce^{-\int p}$; let $C = C(x)$, insert, get $C' = qe^{\int p}$, integrate.

EXAMPLE: $y' - \frac{2}{x}y = x^3$

$p = -\frac{2}{x} \Rightarrow \mu = e^{-2 \ln|x|} = x^{-2}$. Then $(x^{-2}y)' = x \Rightarrow x^{-2}y = \frac{x^2}{2} + C \Rightarrow$

$$y = \frac{x^4}{2} + Cx^2.$$

10.4 Bernoulli and substitutions

► **RECIPE:** $y' + p(x)y = q(x)y^\alpha$, $\alpha \neq 0, 1$

1. Divide by y^α : $y^{-\alpha}y' + py^{1-\alpha} = q$.
2. Set $z := y^{1-\alpha} \Rightarrow z' = (1-\alpha)y^{-\alpha}y'$.
3. $\Rightarrow$ **linear** ODE $z' + (1-\alpha)pz = (1-\alpha)q \Rightarrow$ §10.3.
4. Back-substitute $y = z^{1/(1-\alpha)}$; check the lost solution $y \equiv 0$ separately.

Ex. $y' + y = xy^2$ ($\alpha = \frac{1}{2}$): $z := y^{-1} \Rightarrow z' - z = -x \Rightarrow \mu = e^{-x} \Rightarrow z = x + 1 + Ce^x \Rightarrow y = \frac{1}{x+1+Ce^x}$.

10.5 Constant coefficients: homogeneous part

► **RECIPE:** $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_0y = 0$

1. Ansatz $y = e^{\lambda x} \Rightarrow$ **characteristic polynomial** $\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_0 = 0$.
2. Find all roots (guess an integer root, then polynomial division).
3. Build the fundamental system:
 - simple real root $\lambda \Rightarrow e^{\lambda x}$
 - real root of multiplicity $m \Rightarrow e^{\lambda x}, xe^{\lambda x}, \dots, x^{m-1}e^{\lambda x}$
 - complex pair $\alpha \pm i\beta \Rightarrow e^{\alpha x} \cos \beta x, e^{\alpha x} \sin \beta x$ (multiplicity m : also times $x, \dots, x^{m-1}$)
4. $y_h = C_1y_1 + \dots + C_ny_n$ — exactly n free constants, no more, no fewer.

2nd order quick table ($y'' + by' + cy = 0$, $D = b^2 - 4c$): $D > 0 \Rightarrow C_1e^{\lambda_1x} + C_2e^{\lambda_2x}$; $D = 0 \Rightarrow (C_1 + C_2x)e^{\lambda x}$; $D < 0 \Rightarrow e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x)$.

FACT: Wronskian: is my fundamental system valid?

$$W(x) = \det \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = y_1y_2' - y_1'y_2.$$

$W(x_0) \neq 0$ for one $x_0 \Rightarrow y_1, y_2$ are linearly independent $\Rightarrow$ they span all solutions. For $y'' + by' + cy = 0$: $W(x) = W(0)e^{-bx}$, so W is either never zero or identically zero (Abel).

FACT: reduction of order (one solution known)

Given one solution y_1 of $y'' + py' + qy = 0$, set $y_2 = v(x)y_1$. Inserting kills the v term and leaves a first-order ODE for v' : $v' = \frac{1}{y_1^2}e^{-\int p}$, so $y_2 = y_1 \int \frac{e^{-\int p}}{y_1^2} dx$.

10.6 Particular solution: ansatz by the right-hand side

TABLE: ansatz table — m is the resonance multiplicity

$g(x)$	ansatz y_p
polynomial, degree k	$x^m(A_kx^k + \dots + A_0)$, full polynomial
$Ce^{\mu x}$	$x^m Ae^{\mu x}$
$C \cos \omega x$ or $C \sin \omega x$	$x^m (A \cos \omega x + B \sin \omega x)$, both terms
$P_k(x)e^{\mu x}$	$x^m e^{\mu x} (A_kx^k + \dots + A_0)$
$e^{\mu x} \cos \omega x$	$x^m e^{\mu x} (A \cos \omega x + B \sin \omega x)$
$g_1 + g_2$	solve separately and add (superposition)
$m =$ multiplicity of μ (resp. of $\mu \pm i\omega$) as a root of the characteristic polynomial. No resonance $\Rightarrow m = 0 \Rightarrow$ no extra x .	

► **RECIPE:** execute the ansatz

1. Pick the shape from the table above.
2. **! RESONANCE CHECK:** is the exponent μ (resp. $\pm i\omega$) a root of the characteristic polynomial, and with what multiplicity m ? $\Rightarrow$ multiply the whole ansatz by x^m . Skipping this makes the ansatz collapse to $0 = g$.
3. Differentiate the ansatz, insert into the ODE, compare coefficients, solve for $A, B, \dots$
4. **General solution** $= y_h + y_p$; only now impose the initial conditions, on the *sum*.

EXAMPLE: $y'' - y = e^x$ (textbook resonance)

- 1) $\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1 \Rightarrow y_h = C_1e^x + C_2e^{-x}$.
- 2) $\mu = 1$ is a simple root $\Rightarrow$ resonance, $m = 1 \Rightarrow$ ansatz $y_p = Axe^x$.
- 3) $y_p' = A(1+x)e^x$, $y_p'' = A(2+x)e^x \Rightarrow y_p'' - y_p = 2Ae^x \stackrel{!}{=} e^x \Rightarrow A = \frac{1}{2}$.
- 4) $y = C_1e^x + C_2e^{-x} + \frac{1}{2}xe^x$

EXAMPLE: resonant $\cos$, and a full 2nd-order IVP

(a) $y'' + y = \cos x$: $\lambda = \pm i \Rightarrow y_h = C_1 \cos x + C_2 \sin x$; $\pm i$ is a root $\Rightarrow m = 1 \Rightarrow y_p = x(A \cos x + B \sin x) \Rightarrow A = 0, B = \frac{1}{2} \Rightarrow y = C_1 \cos x + C_2 \sin x + \frac{x}{2} \sin x$.

(b) IVP $y'' - 3y' + 2y = 0$, $y(0) = 1, y'(0) = 0$: $\lambda = 1, 2 \Rightarrow y = C_1e^x + C_2e^{2x}$.
 $C_1 + C_2 = 1, C_1 + 2C_2 = 0 \Rightarrow C_2 = -1, C_1 = 2 \Rightarrow y = 2e^x - e^{2x}$.

FACT: variation of constants (when no ansatz fits, e.g. $g = \frac{1}{x}, \tan x$)

2nd order with $y_h = C_1y_1 + C_2y_2$: set $y_p = c_1(x)y_1 + c_2(x)y_2$ with

$$c_1' = -\frac{y_2g}{W}, \quad c_2' = \frac{y_1g}{W}, \quad W = y_1y_2' - y_1'y_2,$$

then integrate. Always works; just slower than the ansatz.

10.7 Conditions, existence, qualitative behaviour

FACT: IVP vs BVP, Picard–Lindelöf

IVP (*Anfangswertproblem*): all conditions at one point, $y(x_0), y'(x_0), \dots$ — n of them for order n .

BVP (*Randwertproblem*): conditions at two points a, b — may have 0, exactly 1, or infinitely many solutions.

Picard–Lindelöf: $y' = f(x, y), y(x_0) = y_0$ with f continuous and *Lipschitz in y* (e.g. $\partial_y f$ continuous and locally bounded) $\Rightarrow$ a **unique** solution on a neighbourhood of x_0 . Higher order: same after rewriting as a system.

! Continuity of f alone gives existence (Peano) but **not** uniqueness: $y' = \sqrt{|y|}$, $y(0) = 0$ has both $y \equiv 0$ and $y = \frac{x^2}{4}$.

Linear ODEs with continuous coefficients: unique solution on the *whole* interval.

► **RECIPE:** autonomous $y' = f(y)$: sketch without solving

1. Equilibria: the roots of $f(y) = 0$ — these are the constant solutions.
2. Sign of f between consecutive equilibria: $f > 0 \Rightarrow$ solutions increase, $f < 0 \Rightarrow$ decrease.
3. An equilibrium y^* is **stable** if f goes $+$ $\rightarrow$ $-$ there (equivalently $f'(y^*) < 0$), **unstable** if $- \rightarrow +$ ($f'(y^*) > 0$).
4. Solutions can never cross an equilibrium (uniqueness), so each one is trapped between two of them and converges monotonically.

11 Standard proofs from the lecture (Vorlesungsbeweise)

11.1 How to write a proof for marks

► **RECIPE:** generic scheme

1. Write the hypotheses and the goal (“Voraussetzung / Zu zeigen”).
2. Reduce w.l.o.g. (*oBdA*) — e.g. replace f by $-f$ to fix a sign.
3. Build the auxiliary object: a set, a helper function, a subsequence. *Most marks sit here*.
4. Justify that it exists (completeness / Bolzano–Weierstrass / Weierstrass max).
5. Run the ε -argument or the contradiction.
6. Close with the statement and $\square$.

Cheap marks: name every theorem you use and quote its hypotheses (“ f continuous on the *compact* $[a, b]$, hence $\dots$ ”). Writing out the definitions of the given and the wanted object is often already half the marks.

11.2 Tier 1 — expect one of these

FACT: Intermediate value theorem (Zwischenwertsatz) (9 P)

$f \in C([a, b])$, c between $f(a)$ and $f(b) \Rightarrow \exists x^* : f(x^*) = c$.

1. oBdA $f(a) \leq c \leq f(b)$ (else consider $-f$).
2. $X := \{x \in [a, b] \mid f(x) \leq c\}$.
3. $X \neq \emptyset$ ($a \in X$) and $X \subseteq [a, b]$ is bounded $\Rightarrow x^* := \sup X$ exists by completeness.
4. Choose $x_n \in X \cap [x^* - \frac{1}{n}, x^*] \Rightarrow x_n \rightarrow x^*$ with $f(x_n) \leq c$.
5. Sequential continuity $\Rightarrow f(x^*) = \lim f(x_n) \leq c$.
6. Suppose $f(x^*) < c$. Then $x^* < b$ (as $c \leq f(b)$). With $\varepsilon := c - f(x^*) > 0$ continuity gives $\delta > 0$ such that $|y - x^*| < \delta \Rightarrow |f(y) - f(x^*)| < \varepsilon$.
7. Hence $f(y) < f(x^*) + \varepsilon = c$ there, i.e. $(x^*, x^* + \delta) \cap [a, b] \subseteq X$.
8. That contradicts $x^* = \sup X \Rightarrow f(x^*) = c$. $\square$

FACT: Cauchy sequence converges (Vollständigkeit von $\mathbb{R}$) (6 P)

1. Cauchy $\Rightarrow$ **bounded**: with $\varepsilon = 1$ get N , then $|a_n| \leq \max\{|a_1|, \dots, |a_N|, |a_N| + 1\}$.
2. Bolzano–Weierstrass $\Rightarrow$ a convergent subsequence $a_{n_k} \rightarrow L$.
3. Cauchy: $\exists N$ with $|a_n - a_m| < \frac{\varepsilon}{2}$ for all $n, m \geq N$.
4. Subsequence: $\exists N_1$ with $|a_{n_k} - L| < \frac{\varepsilon}{2}$ for all $k \geq N_1$.
5. $N^* := \max\{N, N_1\}$; since $n_k \geq k$, pick one index $n_k \geq N^*$.
6. Triangle inequality: $|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ for all $n \geq N^*$. $\square$

FACT: Mean value theorem (Mittelwertsatz) (3 P)

1. Auxiliary function $g(x) := f(x) - \frac{f(b) - f(a)}{b - a}(x - a)$.
2. $g \in C[a, b]$, differentiable on (a, b) , and $g(a) = f(a) = g(b)$ — the hypotheses of Rolle.
3. Rolle $\Rightarrow \exists \xi \in (a, b)$ with $0 = g'(\xi) = f'(\xi) - \frac{f(b) - f(a)}{b - a}$. $\square$

FACT: Rolle

1. f continuous on the compact $[a, b] \Rightarrow$ attains max and min (Weierstrass).
2. If both are attained at the endpoints, f is constant $\Rightarrow f' \equiv 0$, done.
3. Otherwise an extremum is interior, at some $\xi \Rightarrow$ Fermat $\Rightarrow f'(\xi) = 0$. $\square$

FACT: Bolzano–Weierstrass

Bisection. $(a_n) \subseteq [A, B]$. Halve the interval and keep a half containing infinitely many terms; repeat $\Rightarrow$ nested intervals of length $\frac{B-A}{2^k}$. Pick indices $n_1 < n_2 < \dots$ with a_{n_k} in the k -th interval. The nested-interval principle gives a single point L , and $|a_{n_k} - L| \leq \frac{B-A}{2^k} \rightarrow 0$. $\square$

FACT: monotone + bounded $\Rightarrow$ convergent

(a_n) increasing and bounded $\Rightarrow s := \sup\{a_n\}$ exists (completeness). Given $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound $\Rightarrow \exists N : a_N > s - \varepsilon$. Monotonicity $\Rightarrow s - \varepsilon < a_N \leq a_n \leq s$ for all $n \geq N \Rightarrow |a_n - s| < \varepsilon$. $\square$

11.3 Tier 2 — likely

FACT: extreme value theorem (Weierstrass)

(i) **bounded**: otherwise $\exists x_n$ with $|f(x_n)| \geq n$; Bolzano–Weierstrass $\Rightarrow x_{n_k} \rightarrow x^* \in [a, b]$, continuity $\Rightarrow f(x_{n_k}) \rightarrow f(x^*)$ finite — contradiction.

(ii) **attained**: $S := \sup f$ exists by (i); pick x_n with $f(x_n) \rightarrow S$; a convergent subsequence $x_{n_k} \rightarrow x^*$ gives $f(x^*) = S$ by continuity. $\square$

FACT: continuous on a compact interval $\Rightarrow$ uniformly continuous (Heine)

Suppose not: $\exists \varepsilon_0 > 0$ and x_n, y_n with $|x_n - y_n| < \frac{1}{n}$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0$. Bolzano–Weierstrass $\Rightarrow x_{n_k} \rightarrow x^*$, and then $y_{n_k} \rightarrow x^*$ too. Continuity $\Rightarrow |f(x_{n_k}) - f(y_{n_k})| \rightarrow 0$ — contradiction. $\square$

FACT: fundamental theorem of calculus

$F(x) = \int_a^x f$. Then $\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f = f(\xi_h)$ for some ξ_h between x and $x + h$ (MVT for integrals). As $h \rightarrow 0$, $\xi_h \rightarrow x$, so continuity gives $F'(x) = f(x)$. $\square$

FACT: uniform limit of continuous functions is continuous

The $\frac{\varepsilon}{3}$ -argument: $|f(x) - f(x_0)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)|$. Choose N with $\|f_N - f\|_\infty < \frac{\varepsilon}{3}$ (uniformity: the *same* N for all x — this is the whole point), then δ from continuity of f_N at x_0 . $\square$

FACT: absolute convergence $\Rightarrow$ convergence

Cauchy criterion for series: $|\sum_{k=n}^m a_k| \leq \sum_{k=n}^m |a_k| < \varepsilon$ for $m \geq n \geq N$. So the partial sums are Cauchy, hence convergent. $\square$

FACT: Fermat

x_0 an interior local max: for $h > 0$ small, $\frac{f(x_0+h) - f(x_0)}{h} \leq 0$; for $h < 0$ it is ≥ 0 . Both one-sided limits equal $f'(x_0)$, so $f'(x_0) = 0$. $\square$

FACT: differentiable $\Rightarrow$ continuous

$$f(x) - f(x_0) = \frac{f(x) - f(x_0)}{x - x_0} \cdot (x - x_0) \rightarrow f'(x_0) \cdot 0 = 0. \quad \square$$

11.4 Tier 3 — one-liners you should still know

FACT: convergent $\Rightarrow$ bounded

$\varepsilon = 1$ gives N with $|a_n - a| < 1$ for $n \geq N \Rightarrow |a_n| \leq |a| + 1$; the finitely many earlier terms are bounded by their maximum. $\square$

FACT: convergent $\Rightarrow$ Cauchy
$ a_n - a_m \leq a_n - a + a - a_m < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$ for $n, m \geq N$. $\square$

FACT: the limit is unique
Assume $a_n \rightarrow a$ and $a_n \rightarrow b$ with $a \neq b$. Take $\varepsilon = \frac{ a-b }{2}$: then $ a-b \leq a-a_n + a_n-b < a-b $ — contradiction. $\square$

FACT: squeeze theorem
$b_n \leq a_n \leq c_n$ with $b_n, c_n \rightarrow L$: for large n , $L - \varepsilon < b_n \leq a_n \leq c_n < L + \varepsilon$. $\square$

FACT: limit laws (sum, product)
Sum: $ (a_n + b_n) - (a + b) \leq a_n - a + b_n - b < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$. Product: $ a_n b_n - ab \leq a_n b_n - b + b a_n - a $, and (a_n) is bounded by some C (convergent $\Rightarrow$ bounded) $\Rightarrow \leq C b_n - b + b a_n - a < \varepsilon$. $\square$

FACT: product rule
$\frac{(fg)(x) - (fg)(x_0)}{x - x_0} = f(x) \frac{g(x) - g(x_0)}{x - x_0} + g(x_0) \frac{f(x) - f(x_0)}{x - x_0}$; let $x \rightarrow x_0$ and use $f(x) \rightarrow f(x_0)$. $\square$

FACT: chain rule
Set $\varphi(y) := \frac{g(y) - g(y_0)}{y - y_0}$ for $y \neq y_0$ and $\varphi(y_0) := g'(y_0)$; then φ is continuous at y_0 and $g(y) - g(y_0) = \varphi(y)(y - y_0)$ for all y . Insert $y = f(x)$, $y_0 = f(x_0)$, divide by $x - x_0$, let $x \rightarrow x_0$. $\square$

FACT: derivative of the inverse
f^{-1} continuous and $f'(x_0) \neq 0$: $\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)} \rightarrow \frac{1}{f'(x_0)}$ as $y \rightarrow y_0$ (which forces $x \rightarrow x_0$). $\square$

FACT: geometric series
$s_N = \sum_{k=0}^N q^k$; then $s_N - q s_N = 1 - q^{N+1} \Rightarrow s_N = \frac{1 - q^{N+1}}{1 - q}$. For $ q < 1$, $q^{N+1} \rightarrow 0 \Rightarrow s_N \rightarrow \frac{1}{1 - q}$. $\square$

FACT: root test
$\limsup \sqrt[n]{ a_n } = q < 1$: pick $q < r < 1$; for $n \geq N$, $\sqrt[n]{ a_n } \leq r \Rightarrow a_n \leq r^n \Rightarrow$ comparison with a convergent geometric series. If $q > 1$ then $ a_n \geq 1$ infinitely often $\Rightarrow a_n \not\rightarrow 0$. $\square$

FACT: Cauchy–Hadamard
Apply the root test to $\sum a_n(x - x_0)^n$: $\limsup \sqrt[n]{ a_n } x - x_0 ^n = x - x_0 \limsup \sqrt[n]{ a_n } = \frac{ x - x_0 }{R}$, which is < 1 exactly when $ x - x_0 < R$. $\square$

FACT: comparison test
$0 \leq a_n \leq b_n$ with $\sum b_n < \infty$: the partial sums of $\sum a_n$ are increasing and bounded above by $\sum b_n \Rightarrow$ monotone + bounded $\Rightarrow$ convergent. $\square$

FACT: Leibniz criterion
$b_n \downarrow 0$: (s_{2n}) increases, (s_{2n+1}) decreases, $s_{2n} \leq s_{2n+1}$, and $s_{2n+1} - s_{2n} = b_{2n+1} \rightarrow 0 \Rightarrow$ nested intervals $\Rightarrow$ both converge to a common s . $\square$

FACT: monotone functions are Riemann integrable
f increasing on $[a, b]$, equidistant partition of mesh $\frac{b-a}{n}$: $U(f, P) - L(f, P) = \frac{b-a}{n} \sum_i (f(x_i) - f(x_{i-1})) = \frac{b-a}{n} (f(b) - f(a)) \rightarrow 0$ by telescoping $\Rightarrow$ integrable. $\square$

FACT: uniform convergence allows swapping lim and $\int$
$\left \int_a^b f_n - \int_a^b f \right \leq \int_a^b f_n - f \leq (b - a) \ f_n - f\ _\infty \rightarrow 0$. $\square$

FACT: l'Hospital (idea)
With $f(x_0) = g(x_0) = 0$, the Cauchy MVT gives $\frac{f(x)}{g(x)} = \frac{f(x) - f(x_0)}{g(x) - g(x_0)} = \frac{f'(\xi_x)}{g'(\xi_x)}$ for some ξ_x between x_0 and x ; $x \rightarrow x_0$ forces $\xi_x \rightarrow x_0$. $\square$

FACT: Archimedean property
If $\mathbb{N}$ were bounded above, $s := \sup \mathbb{N}$ would exist; $s - 1$ is not an upper bound $\Rightarrow \exists n > s - 1 \Rightarrow n + 1 > s$, contradicting that s is an upper bound. Hence $\frac{1}{n} \rightarrow 0$. $\square$

FACT: characterisation of the supremum
$s = \sup M \iff$ (i) $x \leq s \forall x \in M$ and (ii) $\forall \varepsilon > 0 \exists x \in M : x > s - \varepsilon$. “ $\Rightarrow$ ”: if (ii) failed for some ε , then $s - \varepsilon$ would be a smaller upper bound. “ $\Leftarrow$ ”: (i) makes s an upper bound, (ii) makes it the least one. $\square$

FACT: Cauchy–Schwarz
For every $t \in \mathbb{R}$: $0 \leq \ x + ty\ ^2 = \ x\ ^2 + 2t\langle x, y \rangle + t^2\ y\ ^2$. A real quadratic in t that is never negative has discriminant ≤ 0 : $4\langle x, y \rangle^2 - 4\ x\ ^2\ y\ ^2 \leq 0$. Equality forces a double root, i.e. $x + ty = 0$ for some t . $\square$

FACT: convex $\iff f'' \geq 0$ (differentiable case)
f convex $\iff$ the chord slopes increase $\iff f'$ is monotonically increasing $\iff f'' \geq 0$ (monotonicity criterion). $\square$

FACT: contraction: uniqueness of the fixed point
$ f(x) - f(y) \leq q x - y $ with $q < 1$. Two fixed points give $ x^* - y^* \leq q x^* - y^* \Rightarrow (1 - q) x^* - y^* \leq 0 \Rightarrow x^* = y^*$. Existence: $x_{n+1} = f(x_n)$ satisfies $ x_{n+1} - x_n \leq q^n x_1 - x_0 $, hence is Cauchy. $\square$

FACT: a convergent sequence has exactly one accumulation point
$a_n \rightarrow a$: every subsequence has indices $n_k \geq k$, so it also converges to a . Hence a is the only subsequential limit. $\square$

12 Exam block: true/false index and model solutions

12.1 True/false scan index

TABLE: sequences and series	
Statement	Reason / counterexample
bounded $\Rightarrow$ convergent	F $(-1)^n$
monotone $\Rightarrow$ convergent	F $a_n = n$
monotone + bounded $\Rightarrow$ convergent	T monotone conv. thm
$ a_{n+1} - a_n \rightarrow 0 \Rightarrow$ convergent	F $\sqrt{n}$, $\sum_{k \leq n} \frac{1}{k}$
Cauchy $\Rightarrow$ convergent (in $\mathbb{R}$)	T completeness
convergent $\Rightarrow$ bounded	T $\varepsilon = 1$ argument
monotone $\Rightarrow$ has an accumulation point	F $a_n = n$ has none
monotone $\Rightarrow$ has ≥ 2 accumulation points	F never more than one
bounded $\Rightarrow$ has an accumulation point	T Bolzano–Weierstrass
min M is an accumulation point	F $x_0 = 0$, $x_n = 1 + \frac{1}{n}$
inf $M = \sup M \Rightarrow$ converges	T the sequence is constant
sup M always exists in $\mathbb{R}$	F unbounded set
$a_n < b_n \forall n \Rightarrow \lim a_n < \lim b_n$	F only $\leq$ survives
$a_n \rightarrow 0 \Rightarrow \sum a_n$ converges	F $\sum \frac{1}{n}$
$\sum a_n$ conv. $\Rightarrow a_n \rightarrow 0$	T divergence test
abs. conv. $\Rightarrow$ conv.	T Cauchy criterion
conv. $\Rightarrow$ abs. conv.	F $\sum \frac{(-1)^n}{n}$
ratio $= 1 \Rightarrow$ diverges	F no information at $q = 1$
$\sum a_n, \sum b_n$ conv. $\Rightarrow \sum a_n b_n$ conv.	F $a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$
reordering never changes the sum	F Riemann, if only conditional

TABLE: functions, continuity, derivatives	
Statement	Reason / counterexample
continuous $\Rightarrow$ differentiable	F $ x $ at 0
differentiable $\Rightarrow$ continuous	T difference quotient
differentiable $\Rightarrow C^1$	F $x^2 \sin \frac{1}{x}$
$f - f(x_0) = O(x - x_0) \Rightarrow$ differentiable	F $ x $ satisfies it
$f - f(x_0) - K(x - x_0) = o(x - x_0) \Rightarrow$ diff.	T that is the definition
one-sided difference quotient exists $\Rightarrow$ diff.	F need both sides
$f'(x_0) = 0 \Rightarrow$ extremum	F x^3 at 0
interior extremum + diff. $\Rightarrow f'(x_0) = 0$	T Fermat
$f''(x_0) = 0 \Rightarrow$ inflection point	F x^4 at 0
continuous on $\mathbb{R} \Rightarrow$ unif. continuous	F $x^2, \sin(x^2)$
continuous on compact $\Rightarrow$ unif. continuous	T Heine
unbounded $f' \Rightarrow$ not unif. continuous	F $\sqrt{x}$ on $[0, 1]$
Lipschitz $\Rightarrow$ unif. continuous	T $\delta = \varepsilon/L$
continuous on $(0, 1] \Rightarrow$ attains its max	F $\frac{1}{x}$, not compact
IVT applies on any interval	F needs continuity + compact
convex, interior critical pt $\Rightarrow$ global min	T f' increasing
convex $\Rightarrow$ global min only on the boundary	F it may be interior
convex $\Rightarrow$ global max on the boundary	T chord above graph
convex diff. $\Rightarrow$ several isolated crit. pts	F critical set is an interval
$\deg p = 5 \Rightarrow 5$ distinct roots in $\mathbb{C}$	F multiplicity (z^5)
$\deg p = 5 \Rightarrow 5$ roots with multiplicity	T fund. thm. of algebra
non-real roots come in conjugate pairs	F only for <i>real</i> coefficients
$\deg p$ odd $\Rightarrow$ a real root	F only for real coefficients

TABLE: integration and function sequences	
Statement	Reason / counterexample
differentiable on $[a, b] \Rightarrow$ integrable	T diff. $\Rightarrow$ cont. $\Rightarrow$ integrable
integrable $\Rightarrow$ no discontinuities	F jumps are fine
integrable $\Rightarrow$ differentiable	F step function
finitely many <i>values</i> $\Rightarrow$ integrable	F Dirichlet $1_{\mathbb{Q}}$
finitely many <i>discontinuities</i> + bdd $\Rightarrow$ integrable	T Lebesgue criterion
monotone on $[a, b] \Rightarrow$ integrable	T telescoping $U - L$
continuous on $[a, b] \Rightarrow$ integrable	T uniform continuity
unbounded f can be Riemann integrable	F that is improper
pointwise $\Rightarrow$ uniform	F x^n on $[0, 1]$
uniform $\Rightarrow$ pointwise	T same N works
pointwise, all f_n cont. $\Rightarrow f$ cont.	F x^n on $[0, 1]$
uniform, all f_n cont. $\Rightarrow f$ cont.	T $\frac{\varepsilon}{3}$ -argument
uniform, f_n cont. $\Rightarrow f$ integrable	T f is continuous
$f_n \Rightarrow f \Rightarrow f'_n \Rightarrow f'$	F $\frac{\sin(nx)}{n}$
pointwise $\Rightarrow \int f_n \rightarrow \int f$	F $n^2 x e^{-nx}$ on $[0, 1]$
$\sup f_n \geq \frac{1}{2n} \Rightarrow$ cannot conv. unif. to 0	F $\frac{1}{2n} \rightarrow 0$ anyway
max at a different x_n each $n \Rightarrow$ not uniform	F only the value matters
$\int_0^\infty f$ conv. $\Rightarrow$ check only ∞	F $\frac{\sin t}{t}$ blows up at 0
$f > 0, \int_1^\infty f < \infty \Rightarrow f \rightarrow 0$	F thin spikes of height 1
$y' = f(x, y), f$ continuous $\Rightarrow$ unique solution	F $y' = \sqrt{ y }, y(0) = 0$

TABLE: power series, Taylor, ODE	
Statement	Reason / counterexample
R is determined by a_n alone	T Cauchy–Hadamard
convergence on $ x - x_0 < R$ is absolute	T root test with $q < 1$
convergence is uniform on $ x - x_0 < R$	F only on closed $[x_0 - r, x_0 + r]$, $r < R$
the two endpoints behave the same	F $\sum \frac{x^n}{n!}$: -1 conv., 1 div.
inside R you may differentiate termwise	T and R is unchanged
$C^\infty \Rightarrow$ equal to its Taylor series	F needs $R_n \rightarrow 0$
Taylor coefficients are unique	T $a_n = \frac{f^{(n)}(x_0)}{n!}$
T_n error $\leq \frac{\max f^{(n+1)} }{(n+1)!} x - x_0 ^{n+1}$	T Lagrange remainder
order n ODE $\Rightarrow n$ free constants	T fundamental system
IVP for a linear ODE always uniquely solvable	T continuous coefficients
BVP always uniquely solvable	F $0, 1$ or ∞ many
μ = root of char. poly. $\Rightarrow$ ansatz needs x^m	T resonance
$g = \cos \omega x \Rightarrow$ ansatz $A \cos \omega x$ suffices	F need $A \cos + B \sin$
separable ODE: constant solutions can be skipped	F they are solutions too
$y' = f(x, y)$, f Lipschitz in $y \Rightarrow$ unique	T Picard–Lindelöf
autonomous: solutions can cross an equilibrium	F uniqueness forbids it

12.2 Exam-day checklist

TRAP: mistakes that cost marks in written proofs

- Using a theorem without checking its hypotheses — above all *compactness* for IVT/Weierstrass/Heine, and *monotonicity* for Leibniz.
- Applying l’Hospital without first verifying the form is $\frac{0}{0}$ or $\frac{\infty}{\infty}$.
- Writing lim of a product/quotient as the product/quotient of limits when one of them does not exist.
- Choosing N or δ that secretly depends on the wrong variable — in a *uniform* statement, δ may not depend on the point.
- Proving one direction of an “ $\Longleftrightarrow$ ” and stopping.
- Concluding “ $<$ ” from a limit: $a_n < b_n$ only gives $\lim a_n \leq \lim b_n$.
- Dividing by something that can vanish (e.g. $g(y)$ in a separable ODE) without treating that case.
- Forgetting the constant solutions, the $+C$, or the second endpoint of an improper integral.
- Asserting a sup/max exists without invoking completeness or Weierstrass.
- Doing algebra instead of stating the argument: name the theorem, that is where the marks are.

TRAP: read this in the first minute

- Integral asked, options given?** Differentiate the options instead of integrating.
- Limit as $x \rightarrow 0$?** Taylor beats l’Hospital almost always.
- Improper integral?** Split first, check *both* endpoints.
- Separable ODE?** Write down the constant solutions $g(y) = 0$.
- Forced linear ODE?** Resonance check *before* the ansatz.
- “Zeigen Sie, dass es konvergiert”?** Name the theorem (monotone + bounded / Cauchy / comparison). Naming earns marks.
- “Begründen Sie”?** One sentence per step, and quote the hypotheses of every theorem you cite.
- Substitution finished?** Differentiate the result; the chain-rule factor must reappear.
- Uniform convergence?** Compute $\sup |f_n - f|$ explicitly; a maximiser x_n is already a complete answer for “no”.
- Stuck on a proof?** Write out the definition of everything given and everything to be shown. Often half the marks.
- Multiple choice, no negative marking:** never leave a box empty.

12.3 Model solutions for the open questions

EXAMPLE: “Untersuchen Sie $f_n(x) = \frac{x}{1+nx^2}$ auf $\mathbb{R}$ ” (9 P)

a) Gleichmässige Stetigkeit. For $n = 0$, $f_0(x) = x$ is Lipschitz with constant 1, hence uniformly continuous. For $n \geq 1$, f_n is differentiable on $\mathbb{R}$ with $f'_n(x) = \frac{1-nx^2}{(1+nx^2)^2}$, so $|f'_n(x)| \leq \frac{1+nx^2}{(1+nx^2)^2} = \frac{1}{1+nx^2} \leq 1$. By the mean value theorem $|f_n(x) - f_n(y)| \leq |x - y|$ for all x, y , so $\delta := \varepsilon$ satisfies the definition. Hence *every* f_n is uniformly continuous.

b) Punktwaiser Grenzwert. Fix x . For $x = 0$, $f_n(0) = 0$ for all n . For $x \neq 0$, $|f_n(x)| = \frac{|x|}{1+nx^2} \leq \frac{|x|}{nx^2} = \frac{1}{n|x|} \rightarrow 0$. Hence $f_n \rightarrow f \equiv 0$ pointwise on $\mathbb{R}$.

c) Gleichmässige Konvergenz. $\|f_n - f\|_\infty = \sup_x |f_n(x)|$. From $f'_n(x) = 0$ we get $x = \pm \frac{1}{\sqrt{n}}$, with $f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1}{2\sqrt{n}}$, and $f_n(x) \rightarrow 0$ as $|x| \rightarrow \infty$. Therefore $\|f_n\|_\infty = \frac{1}{2\sqrt{n}} \rightarrow 0$, i.e. $f_n \Rightarrow 0$ **uniformly on all of** $\mathbb{R}$. $\square$

EXAMPLE: “ $f(x) = \frac{\sin x}{1+x}$ ”: Fixpunkt und Rekursion” (9 P)

a) Unique fixed point, and $|f(x)| < |x|$ for $x \neq 0$. $f(0) = 0$, so 0 is a fixed point. For $x \neq 0$ use the strict inequality $|\sin x| < |x|$ together with $1 + x^2 \geq 1$:

$$|f(x)| = \frac{|\sin x|}{1+x^2} \leq |\sin x| < |x|.$$

If $x^* \neq 0$ were another fixed point then $|x^*| = |f(x^*)| < |x^*|$, a contradiction. So $x^* = 0$ is the **unique** fixed point.

b) Behaviour of $x_{n+1} = f(x_n)$, $x_0 \in (0, 1)$. *Positivity:* for $x \in (0, 1)$ we have $\sin x > 0$, so $f(x) \in (0, x) \subseteq (0, 1)$; by induction $x_n \in (0, 1)$ for all n . *Monotonicity:* by part a), $x_{n+1} = f(x_n) < x_n$, so (x_n) is strictly decreasing. *Conclusion:* decreasing and bounded below by 0, hence convergent by the monotone convergence theorem; let $L := \lim x_n \geq 0$. Since f is continuous, $L = \lim x_{n+1} = \lim f(x_n) = f(L)$, so L is a fixed point, and by a) $L = 0$. Thus $x_n \rightarrow 0$ for every $x_0 \in (0, 1)$. $\square$

EXAMPLE: box answers (3 P each, only the result is marked)

Limit. $\lim_{x \rightarrow 0} \frac{\sin x - x + x^3/6}{x^5}$: insert $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \Rightarrow$ numerator $= \frac{x^5}{120} + O(x^7) \Rightarrow \frac{1}{120}$.

ODE. $y'' - y = e^x$: $\lambda = \pm 1$; $\mu = 1$ is a simple root $\Rightarrow$ resonance $\Rightarrow y_p = A x e^x$, $y_p'' - y_p = 2A e^x \Rightarrow A = \frac{1}{2} \Rightarrow y = C_1 e^x + C_2 e^{-x} + \frac{1}{2} x e^x$.

Integral. $\int \frac{dx}{1+\cos x}$: $1 + \cos x = 2 \cos^2 \frac{x}{2}$, then $u = \frac{x}{2}$, $dx = 2 du \Rightarrow \int \frac{du}{\cos^2 u} = \tan u \Rightarrow \tan \frac{x}{2} + C$.

Radius. $\sum \frac{n^n (x-2)^n}{(2n)!}$: coefficient ratio $\rightarrow 0 \Rightarrow R = \infty \Rightarrow$ converges for all $x \in \mathbb{R}$.

Improper. $\int_1^\infty \frac{\cos t}{t} dt$ conv.; $\int_1^\infty \frac{\cos^2 t}{t} dt$ div.; $\int_0^\infty \frac{\sin t}{t^3} dt$ div. (at $t = 0$).

EXAMPLE: “Für welche x konvergiert die Reihe?” (radius + endpoints)

For $\sum_{n \geq 1} \frac{(x-3)^n}{n 2^n}$: the centre is $x_0 = 3$ and $a_n = \frac{1}{n 2^n}$.

Radius: $\left| \frac{a_{n+1}}{a_n} \right| = \frac{n 2^n}{(n+1) 2^{n+1}} = \frac{n}{2(n+1)} \rightarrow \frac{1}{2}$, hence $R = 2$. By Cauchy–Hadamard the series converges absolutely for $|x - 3| < 2$ and diverges for $|x - 3| > 2$.

Endpoint $x = 5$: the series becomes $\sum \frac{2^n}{n 2^n} = \sum \frac{1}{n}$, the harmonic series, which **diverges**.

Endpoint $x = 1$: it becomes $\sum \frac{(-2)^n}{n 2^n} = \sum \frac{(-1)^n}{n}$, which **converges** by the Leibniz criterion ($\frac{1}{n}$ decreases monotonically to 0), though not absolutely. *Answer:* the series converges exactly for $x \in [1, 5)$. $\square$

! Always test the two endpoints *separately*; they need not agree.

EXAMPLE: “Konvergiert das uneigentliche Integral?” (write-up)

For $\int_0^\infty \frac{dx}{\sqrt{x}(1+x)}$ there are *two* problem points, so we split: $\int_0^\infty = \int_0^1 + \int_1^\infty$, and both pieces must converge.

At $x \rightarrow 0^+$: $\frac{1}{\sqrt{x}(1+x)} \sim \frac{1}{\sqrt{x}} = x^{-1/2}$, and $\int_0^1 x^{-p} dx$ converges for $p < 1$; here $p = \frac{1}{2} < 1$, so $\int_0^1$ **converges**.

At $x \rightarrow \infty$: $\frac{1}{\sqrt{x}(1+x)} \sim \frac{1}{x^{3/2}}$, and $\int_1^\infty x^{-p} dx$ converges for $p > 1$; here $p = \frac{3}{2} > 1$, so $\int_1^\infty$ **converges**.

Both pieces converge, hence the integral converges. (Its value is π , via $x = t^2$.) $\square$

! The exponent threshold *reverses* between the two ends: mild singularity allowed at 0, fast decay required at ∞ .

EXAMPLE: “Bestimmen Sie Extrema und Wendepunkte” (curve discussion write-up)

For $f(x) = \frac{x}{1+x^2}$: *Domain* $\mathbb{R}$, f is odd and continuous as a quotient of continuous functions with denominator $\neq 0$. *First derivative* $f'(x) = \frac{1-x^2}{(1+x^2)^2}$, so $f'(x) = 0 \iff x = \pm 1$. *Second derivative* $f''(x) = \frac{2x(x^2-3)}{(1+x^2)^3}$; $f''(1) = -\frac{1}{2} < 0$, hence $x = 1$ is a local maximum with $f(1) = \frac{1}{2}$, and $f''(-1) = \frac{1}{2} > 0$, hence $x = -1$ is a local minimum with $f(-1) = -\frac{1}{2}$. *Monotonicity* $f' > 0$ on $(-1, 1)$ and $f' < 0$ outside, so f increases on $[-1, 1]$ and decreases on $(-\infty, -1]$ and $[1, \infty)$. *Inflection points* f'' changes sign at $x = 0, \pm\sqrt{3}$. *Asymptote* $\lim_{x \rightarrow \pm\infty} f(x) = 0$, so $y = 0$. Since $f \rightarrow 0$ at $\pm\infty$ and the only candidates are ± 1 , the local extrema are also *global*: $\max f = \frac{1}{2}$, $\min f = -\frac{1}{2}$, both attained. $\square$

13 Lookup tables

13.1 How to use the derivative / antiderivative table

TRAP: read this before you copy a single row

Every row means $\frac{d}{dx} F(x) = f(x)$ and $\int f(x) dx = F(x) + C$, written for the **bare argument** x .

- Inner function linear,** $ax + b$: $\int f(ax + b) dx = \frac{1}{a} F(ax + b) + C$ — divide by the inner derivative a .
 - Inner function anything else:** you may **not** patch with a factor. Substitute $u = g(x)$, $dx = \frac{dx}{g'(x)}$, and continue only if the leftover $g'(x)$ cancels.
 - Constant prefactors** stay outside: $\int c f = c \int f$. Do the prefactor and the substitution factor as *separate written steps*; never patch by feeling.
 - Differentiate your answer.** The chain-rule factor must reappear.
- The trap in full: $\int \frac{dx}{2 \cos^2(x/2)}$. With $u = \frac{x}{2}$, $dx = 2 du$: $\int \frac{2 du}{2 \cos^2 u} = \tan u = \tan \frac{x}{2} + C$ — the 2 from dx cancels the $\frac{1}{2}$ in front, so the answer is **not** $2 \tan \frac{x}{2}$.

13.2 Derivatives and antiderivatives

TABLE: powers, roots, exponentials, logarithms

$F = \int f$	$ f(x) $	$ f'(x) $
$\frac{x^{n+1}}{n+1}$ ($n \neq -1$)	x^n	$n x^{n-1}$
$\ln x $	$\frac{1}{x}$	$-\frac{1}{x^2}$
$\frac{2}{3} x^{3/2}$	$\frac{1}{\sqrt{x}}$	$-\frac{1}{2\sqrt{x}}$
$\frac{-n}{n+1} x^{(n+1)/n}$	$\sqrt[n]{x}$	$\frac{1}{n} x^{1/n-1}$
$\frac{a(x+b)^{n+1}}{a(n+1)}$	$(ax+b)^n$	$an(ax+b)^{n-1}$
e^x	e^x	e^x
$\frac{1}{a} e^{ax}$	e^{ax}	$a e^{ax}$
$\frac{a^x}{\ln a}$	a^x	$a^x \ln a$
$x(\ln x - 1)$	$\ln x $	$\frac{1}{x}$
$\frac{x}{\ln a} (\ln x - 1)$	$\log_a x $	$\frac{1}{x \ln a}$

TABLE: trigonometric and inverse trigonometric		
$F = \int f$	$f(x)$	$f'(x)$
$-\cos x$	$\sin x$	$\cos x$
$\sin x$	$\cos x$	$-\sin x$
$-\ln \cos x $	$\tan x$	$\frac{1}{\cos^2 x} = 1 + \tan^2 x$
$\ln \sin x $	$\cot x$	$-\frac{1}{\sin^2 x}$
$\ln\left \tan\frac{x}{2}\right $	$\frac{1}{\sin x}$	$-\frac{\cos x}{\sin^2 x}$
$\ln\left \tan\left(\frac{x}{2} + \frac{\pi}{4}\right)\right $	$\frac{1}{\cos x}$	$\frac{\sin x}{\cos^2 x}$
$\frac{x}{2} - \frac{\sin 2x}{4}$	$\sin^2 x$	$\sin 2x$
$\frac{x}{2} + \frac{\sin 2x}{4}$	$\cos^2 x$	$-\sin 2x$
$x \arcsin x + \sqrt{1-x^2}$	$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$
$x \arccos x - \sqrt{1-x^2}$	$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$
$x \arctan x - \frac{1}{2} \ln(1+x^2)$	$\arctan x$	$\frac{1}{1+x^2}$

TABLE: hyperbolic		
$F = \int f$	$f(x)$	$f'(x)$
$\cosh x$	$\sinh x$	$\cosh x$
$\sinh x$	$\cosh x$	$\sinh x$
$\ln \cosh x$	$\tanh x$	$\frac{1}{\cosh^2 x}$
$x \operatorname{arsinh} x - \sqrt{1+x^2}$	$\operatorname{arsinh} x$	$\frac{1}{\sqrt{1+x^2}}$
$x \operatorname{arcosh} x - \sqrt{x^2-1}$	$\operatorname{arcosh} x$	$\frac{1}{\sqrt{x^2-1}}$
$x \operatorname{artanh} x + \frac{1}{2} \ln(1-x^2)$	$\operatorname{artanh} x$	$\frac{1}{1-x^2}$
$\sinh x = \frac{e^x - e^{-x}}{2}$, $\cosh x = \frac{e^x + e^{-x}}{2}$, $\cosh^2 - \sinh^2 = 1$; $\operatorname{arsinh} x = \ln(x + \sqrt{x^2+1})$, $\operatorname{arcosh} x = \ln(x + \sqrt{x^2-1})$, $\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$.		

TABLE: antiderivatives you will not guess (1)	
$f(x)$	$F(x) = \int f \, dx$
$\frac{1}{ax+b}$	$\frac{1}{a} \ln ax+b $
$\frac{f'(x)}{f(x)}$	$\ln f(x) $
$\frac{1}{a^2+x^2}$	$\frac{1}{a} \arctan \frac{x}{a}$
$\frac{1}{a^2-x^2}$	$\frac{1}{2a} \ln \left \frac{a+x}{a-x} \right $
$\frac{1}{\sqrt{a^2-x^2}}$	$\arcsin \frac{x}{ a }$
$\frac{1}{\sqrt{x^2 \pm a^2}}$	$\ln x + \sqrt{x^2 \pm a^2} $
$\ln x$	$x \ln x - x$
$\frac{1}{1+\cos x}$	$\tan \frac{x}{2}$
$\frac{1}{1-\cos x}$	$-\cot \frac{x}{2}$
$\frac{1}{\sin x \cos x}$	$\ln \tan x $

TABLE: antiderivatives you will not guess (2)	
$f(x)$	$F(x) = \int f \, dx$
$\sqrt{a^2-x^2}$	$\frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \arcsin \frac{x}{ a }$
$\sqrt{a^2+x^2}$	$\frac{x}{2} \sqrt{a^2+x^2} + \frac{a^2}{2} \ln x + \sqrt{a^2+x^2} $
$\sqrt{x^2-a^2}$	$\frac{x}{2} \sqrt{x^2-a^2} - \frac{a^2}{2} \ln x + \sqrt{x^2-a^2} $
xe^{ax}	$\frac{e^{ax}}{a^2} (ax-1)$
$e^{ax} \sin bx$	$\frac{e^{ax}}{a^2+b^2} (a \sin bx - b \cos bx)$
$e^{ax} \cos bx$	$\frac{e^{ax}}{a^2+b^2} (a \cos bx + b \sin bx)$
$x \sin x$	$\sin x - x \cos x$
$x \cos x$	$\cos x + x \sin x$
x^x	no elementary form; $\frac{d}{dx} x^x = x^x(1 + \ln x)$

13.3 Limits, growth, series

FACT: growth hierarchy — kills most limits instantly	
As $x \rightarrow \infty$, for $\alpha > 0$ and $b > 1$:	
$\ln x \ll x^\alpha \ll b^x \ll x! \ll x^x$	
Every ratio of a term on the left to one on the right tends to 0.	

TABLE: standard limits	
$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$	$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$	$\lim_{x \rightarrow 0} \frac{\arctan x}{x} = 1$
$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$	$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$
$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$	$\lim_{x \rightarrow 1} \frac{\ln x}{x-1} = 1$
$\lim_{x \rightarrow 0^+} x \ln x = 0$	$\lim_{x \rightarrow 0^+} x^x = 1$
$\lim_{x \rightarrow \infty} \frac{\ln x}{x^n} = 0$	$\lim_{x \rightarrow \infty} \frac{x^n}{e^x} = 0$
$\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$	$\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1 \ (a > 0)$
$\lim_{n \rightarrow \infty} \frac{n!}{n!} = 0$	$\lim_{n \rightarrow \infty} \sqrt[n]{n!} = \infty$
$\lim_{x \rightarrow 0} (1+x)^{1/x} = e$	$\lim_{x \rightarrow \pm\infty} \left(1 + \frac{k}{x}\right)^{mx} = e^{km}$
$\lim_{x \rightarrow \infty} x^n q^x = 0 \ (q < 1)$	$\lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}$

TABLE: Taylor series at 0, with radius	
$e^x = \sum_{n \geq 0} \frac{x^n}{n!}$, $R = \infty$ $\sin x = \sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$, $R = \infty$ $\cos x = \sum_{n \geq 0} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$, $R = \infty$ $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$, $R = \frac{\pi}{2}$ $\frac{1}{1-x} = \sum_{n \geq 0} x^n$, $R = 1$ $\ln(1+x) = \sum_{n \geq 1} \frac{(-1)^{n+1} x^n}{n}$, $R = 1$ $\arctan x = \sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{2n+1}$, $R = 1$ $(1+x)^a = \sum_{n \geq 0} \binom{a}{n} x^n$, $R = 1$ $\sinh x = \sum_{n \geq 0} \frac{x^{2n+1}}{(2n+1)!}$, $\cosh x = \sum_{n \geq 0} \frac{x^{2n}}{(2n)!}$, $R = \infty$	

13.4 Trigonometry, logs, algebra

TABLE: trigonometric identities	
$\sin^2 x + \cos^2 x = 1$; $1 + \tan^2 x = \frac{1}{\cos^2 x}$ $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ $\sin 2x = 2 \sin x \cos x$; $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x$ $\sin^2 x = \frac{1 - \cos 2x}{2}$; $\cos^2 x = \frac{1 + \cos 2x}{2}$ $1 + \cos x = 2 \cos^2 \frac{x}{2}$; $1 - \cos x = 2 \sin^2 \frac{x}{2}$ $\tan \frac{x}{2} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$ $\sin(-x) = -\sin x$ (odd), $\cos(-x) = \cos x$ (even) $\sin(\pi - x) = \sin x$, $\cos(\pi - x) = -\cos x$ $\sin\left(x + \frac{\pi}{2}\right) = \cos x$, $\cos\left(x + \frac{\pi}{2}\right) = -\sin x$ $\arcsin: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $\arccos: [-1, 1] \rightarrow [0, \pi]$, $\arctan: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	
Forgot one? Derive it from e^{ix} (§2.1) in ten seconds.	

TABLE: special values and quadrant signs				
deg	rad	sin	cos	tan
0	0	0	1	0
30	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90	$\frac{\pi}{2}$	1	0	—
120	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\sqrt{3}$
180	π	0	-1	0

Quad.	sin	cos	tan
I	+	+	+
II	+	-	-
III	-	-	+
IV	-	+	-

Solving $\sin x = c$: $x = \arcsin c + 2k\pi$ and $x = \pi - \arcsin c + 2k\pi$.
 $\cos x = c$: $x = \pm \arccos c + 2k\pi$. $\tan x = c$: $x = \arctan c + k\pi$.

TABLE: logs, powers, algebra survival kit	
$\log_a x = \frac{\ln x}{\ln a}$; $a^x = e^{x \ln a}$; $u^v = e^{v \ln u}$ $\ln(xy) = \ln x + \ln y$; $\ln \frac{x}{y} = \ln x - \ln y$; $\ln(x^y) = y \ln x$ $\ln 1 = 0$, $\ln e = 1$; $\ln x$ defined only for $x > 0$ $\sqrt{A} - \sqrt{B} = \frac{A-B}{\sqrt{A} + \sqrt{B}}$ (conjugate trick, kills $\infty - \infty$) $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$ - complete the square: $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$ $\binom{n}{k} = \frac{n!}{k!(n-k)!}$; $(a+b)^n = \sum_k \binom{n}{k} a^k b^{n-k}$ $\sum_{k=1}^n k = \frac{n(n+1)}{2}$; $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$; $\sum_{k=0}^n q^k = \frac{1-q^{n+1}}{1-q}$ - guess a polynomial root among the divisors of the constant term, then divide	

13.5 Decision tables and glossary

TABLE: I need to... → use	
show a sequence converges without knowing the limit	monotone + bounded, or Cauchy
show a value is attained	IVT, or Weierstrass max/min
show a zero or fixed point exists	IVT applied to $f(x) - x$
show it is unique	strict monotonicity, or Rolle, or contraction
classify a critical point	sign of f'' ; if $f'' = 0$, sign change of f'
evaluate $\frac{0}{0}$ or $\frac{\infty}{\infty}$	Taylor (preferred) or l'Hospital
handle $0 \cdot \infty$, $\infty - \infty$, 1^∞	rewrite first, then l'Hospital
decide f^∞ f converges	compare with x^{-p} , check <i>both</i> ends
integrate $\sqrt{a^2 \pm x^2}$	trigonometric substitution
integrate a rational function	partial fractions
integrate rational in $\sin, \cos$	half-angle, else $t = \tan \frac{x}{2}$
solve a forced linear ODE	$y_h + y_p$, resonance check
solve a 1st order nonlinear ODE	separation / $u = y/x$ / Bernoulli / int. factor
show $f_n \Rightarrow f$	$\sup f_n - f \rightarrow 0$
disprove $f_n \Rightarrow f$	one x_n with $ f_n(x_n) - f(x_n) \geq \varepsilon_0$
swap lim and $\int$	needs <i>uniform</i> convergence
prove an inequality	MVT, convexity/tangent line, or induction

TABLE: series looks like... → take	
$a_n \nearrow 0$	divergence test, done
$\sum \frac{1}{n^p}$, $\sum q^n$	p -series / geometric
factorials	ratio test
n -th powers	root test
alternating with $b_n \downarrow 0$	Leibniz
resembles a known series	comparison on leading powers
$a_n = f(n)$ with $f \downarrow$	integral test
a_n, b_n , $b_n \downarrow 0$, $\sum a_n$ bounded	Abel/Dirichlet
power series	Cauchy–Hadamard + endpoints

TABLE: word list DE → EN	
Folge = sequence • Reihe = series • Grenzwert = limit • Häufungspunkt = accumulation point • beschränkt = bounded • monoton = monotone • Teilfolge = subsequence • Schranke = bound • stetig = continuous • gleichmäßig = uniform(ly) • punktweise = pointwise • differenzierbar = differentiable • Ableitung = derivative • Stammfunktion = antiderivative • Zwischenwertsatz = IVT • Mittelwertsatz = MVT • Umkehrfunktion = inverse • Nullstelle = root, zero • Wendepunkt = inflection point • Extremstelle = extremum • Abschätzung = estimate • oBdA = w.l.o.g. • Voraussetzung = hypothesis • Behauptung = claim • Widerspruch = contradiction • Anfangswertproblem = IVP • Randwertproblem = BVP • Trennung der Variablen = separation of variables • uneigentlich = improper • Konvergenzradius = radius of convergence • Partialsumme = partial sum • Partialbruchzerlegung = partial fractions • Kettenregel = chain rule • Produktregel = product rule • konvex = convex • Treppenfunktion = step function • Zerlegung = partition • Obersumme/Untersumme = upper/lower sum.	